

**DSCC**

Cloud Computing — Project Report



**Cloud-Native Resume Screener & Candidate Pipeline**

*Submitted by*

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**Live Project:** <http://16.170.208.202:5000/>

**GitHub Repository:** [github.com/shravanbpatel954/CloudATS-CareerPortal](https://github.com/shravanbpatel954/CloudATS-CareerPortal)

# 1. Executive Summary & Project Overview

CloudATS is an enterprise-grade, cloud-native Applicant Tracking System (ATS) and Candidate Intelligence Screener. Built on modern web standards (React 18, Tailwind CSS, Express 5, and Node.js), CloudATS automates the resume screening and candidate evaluation lifecycle through deep integration with Amazon Web Services (AWS).

## Key Objectives

- Zero-Friction Candidate Application Portal: public-facing, distraction-free application links (/?apply=<jobId>) allowing candidates to upload resumes and submit structured profiles.
- Automated Resume Ingestion & OCR: multi-format document parsing (.pdf, .docx, .doc, .txt, .png, .jpg) backed by AWS document intelligence.
- Intelligent ATS Scoring Engine: weighted multi-factor candidate scoring (Skills Match, Experience Ratio, Document Keyword Relevance) with automatic leaderboard ranking (#1 Top Match, #2 Good Match, #3+ Moderate).
- Recruiter Productivity Suite: 1-click job role creation, custom skill taxonomy assignment, AI-generated technical interview questions, encrypted S3 presigned document viewing, and 1-click Excel/CSV pipeline export.
- Enterprise Cloud Observability: real-time AWS telemetry, CloudWatch custom metric publishing, and automated SES candidate email confirmations.

# 2. AWS Cloud Architecture

CloudATS follows a single-server, cloud-integrated architecture. The Node.js/Express backend, running on an Amazon EC2 instance, serves the React single-page application and orchestrates all AWS service calls. The high-level request flow is as follows:

- 1. The Candidate Application Portal (public, /?apply=job-id) and the Recruiter ATS Command Center (passcode protected) both submit requests to the EC2-hosted backend.
- 2. The backend authenticates its AWS credentials via AWS STS.
- 3. Uploaded resumes are stored in an Amazon S3 bucket, partitioned by job ID.
- 4. Documents are sent to Amazon Textract for OCR-based text extraction.
- 5. Image-based uploads (certificates, credentials) are analyzed by Amazon Rekognition.
- 6. If Textract is unavailable, a local Mammoth/PDF-Parse fallback engine extracts text.
- 7. Application and screening events publish custom telemetry to Amazon CloudWatch.
- 8. Amazon SES dispatches an automated confirmation email to the candidate, and S3 generates a time-limited, presigned download link for recruiters.

## Layers of the Architecture

| Layer        | Components                                                                               |
|--------------|------------------------------------------------------------------------------------------|
| Client Layer | Candidate Application Portal (public); Recruiter ATS Command Center (passcode-protected) |

| Layer                         | Components                                                                                                             |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Compute & Hosting             | Amazon EC2 (Ubuntu 24.04 LTS), Node.js 22 + Express 5 backend, React 18 SPA, PM2 process manager (port 5000)           |
| Storage & Identity            | Amazon S3 (partitioned resume storage, presigned URLs), AWS STS (session/identity), AWS IAM (least-privilege policies) |
| AI & Document Intelligence    | Amazon Textract (OCR), Amazon Rekognition (image/label analysis), local Mammoth/PDF-Parse fallback engine              |
| Observability & Communication | Amazon CloudWatch (custom metrics), Amazon SES (candidate email receipts)                                              |

### 3. Deep-Dive: AWS Services Utilized

| AWS Service        | Core Purpose                         | Integration Details                                                                                  |
|--------------------|--------------------------------------|------------------------------------------------------------------------------------------------------|
| Amazon EC2         | Production application hosting       | Hosts the unified Node.js runtime and statically served React SPA under the PM2 process supervisor.  |
| Amazon S3          | Object storage for resumes           | Secure partitioned bucket storage (/resumes/{jobId}/...) with encrypted presigned URLs.              |
| Amazon Textract    | AI document OCR & parsing            | Optical character recognition for extracting text and structured data from PDF resumes.              |
| Amazon Rekognition | Image analysis & verification        | Deep-learning visual inspection, credential badge recognition, and label detection.                  |
| Amazon CloudWatch  | Operational metrics & monitoring     | Custom metric publishing under namespace CloudATS/RecruitmentPipeline for real-time telemetry.       |
| Amazon SES         | Transactional email dispatch         | Sends automated, branded application confirmation emails to candidates upon successful submission.   |
| AWS STS            | Identity & session validation        | Dynamically verifies IAM caller identity and supports short-lived session tokens (AWS Academy/Labs). |
| AWS IAM            | Access control & security governance | Enforces least-privilege access policies for backend services.                                       |

#### 3.1 Amazon EC2 (Elastic Compute Cloud)

- Role: serves as the compute backbone hosting the entire CloudATS platform.
- OS: Ubuntu Linux (24.04 LTS).
- Runtime: Node.js 22.x LTS with npm 10.x.
- Process Management: PM2 ensures continuous execution, background uptime, automatic reboot recovery, and memory monitoring.
- Networking & Firewall: EC2 Security Group configured with an inbound TCP rule on port 5000.
- Single-Server Architecture: the Express 5 backend directly serves the pre-compiled production Vite/React build from /dist, minimizing compute footprint and latency.

#### 3.2 Amazon S3 (Simple Storage Service)

Highly scalable, durable (99.999999999% durability) object store for resumes and applicant documents.

```
s3://your-cloudats-bucket/resumes/{jobId}/{timestamp}_{filename}
```

- Presigned URLs (@aws-sdk/s3-request-presigner) generate temporary, cryptographically signed URLs for viewing or downloading candidate resumes without making the bucket public.

- Selectable expiration windows: 1 Hour, 24 Hours, or 7 Days.
- Metadata tagging stores candidate name, email, target job role ID, and application timestamp.

### 3.3 Amazon Textract

- DetectDocumentTextCommand analyzes raw document byte arrays to extract high-accuracy text from multi-column PDF resumes.
- Hybrid fallback engine: if Textract hits rate limits or non-standard formats, CloudATS falls back to embedded multi-format parsers (pdf-parse, mammoth for .docx, and text sanitizers).

### 3.4 Amazon Rekognition

- DetectLabelsCommand automatically analyzes image-based candidate uploads (certificates, portfolios, image resumes), identifying technical objects, document types, and visual properties with confidence percentages (>70%).
- DetectTextCommand extracts embedded text from graphical images and scanned certificates.

### 3.5 Amazon CloudWatch

- Namespace: CloudATS/RecruitmentPipeline; Dimension: Environment=Production.
- Key metrics tracked: ResumeUploaded (Count), CandidateEvaluated (Count), PipelineExported (Count).
- Executed asynchronously using PutMetricDataCommand to ensure zero impact on candidate response times.

### 3.6 Amazon SES (Simple Email Service)

- Dispatches immediate, HTML-formatted confirmation receipts to candidates upon resume upload.
- Includes candidate application ID (app-xxxxxx), submitted job title, and company receipt acknowledgment.
- Handles sandbox verification gracefully with informative console logging when SES is in development mode.

### 3.7 AWS STS & IAM

- GetCallerIdentityCommand validates active IAM credentials, AWS Account ID, and ARN on server startup.
- Full support for AWS\_SESSION\_TOKEN for temporary session-based credential rotation (AWS Academy / Learner Lab).

Least-Privilege IAM Policy used for this project:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": ["s3:PutObject", "s3:GetObject", "s3:ListBucket", "s3:DeleteObject"],

```

```
    "Resource": [
      "arn:aws:s3:::your-cloudats-bucket",
      "arn:aws:s3:::your-cloudats-bucket/*"
    ],
  },
  {
    "Effect": "Allow",
    "Action": [
      "textract:DetectDocumentText",
      "rekognition:DetectLabels",
      "rekognition:DetectText",
      "cloudwatch:PutMetricData",
      "ses:SendEmail",
      "sts:GetCallerIdentity"
    ],
    "Resource": "*"
  }
]
```

## 4. ATS Scoring Algorithm & Candidate Ranking Logic

CloudATS evaluates applicant resumes against role requirements using a multi-dimensional weighted matching algorithm:

**ATS Total Score = (0.55 × Skills Score) + (0.25 × Experience Score) + (0.20 × Relevance Score)**

### 1. Skills Score (55% Weight)

- Matches parsed resume text against Required Skills (weighted 1.0×) and Preferred Skills (bonus weight 0.5×).
- Compares against a comprehensive Tech Skills Dictionary spanning 100+ technologies across Testing/QA, Frontend, Backend, Cloud/DevOps, Databases, and AI/ML.

### 2. Experience Score (25% Weight)

Evaluates candidate years of experience against the role requirement:

*Experience Score = min( 100, (Candidate Years / Required Years) × 100 )*

### 3. Relevance Score (20% Weight)

Analyzes term frequency-inverse density, keyword density, and role-specific taxonomy occurrences across the extracted resume text.

### 4. Automated Ranking & Tier Badges

| Rank              | Criteria        | Badge                                  |
|-------------------|-----------------|----------------------------------------|
| #1 — Strong Match | Score ≥ 80%     | Emerald highlight + leaderboard crown  |
| #2 — Good Match   | Score 60% – 79% | Blue badge + recommended for interview |

| Rank               | Criteria    | Badge                                    |
|--------------------|-------------|------------------------------------------|
| #3+ — Moderate Fit | Score < 60% | Amber badge + skill gap review suggested |

## 5. 1-Click Excel / CSV Export Feature

Recruiters can export candidate evaluation pipelines for any role with a single click. Files are encoded in UTF-8 with a Byte Order Mark (\uFEFF), ensuring full compatibility with Microsoft Excel, Apple Numbers, and Google Sheets. The exported data fields include:

- Rank Position (#1, #2, ...)
- Candidate Name
- Email & Phone Number
- Years of Experience
- Target Job Role Title
- ATS Overall Match Score (%)
- Fit Recommendation (Top Match, Good Match, Moderate Fit)
- Matched Skills List
- Missing Skills / Gaps
- Skills Score breakdown (%)
- Experience Score breakdown (%)
- Keyword Relevance Score breakdown (%)
- S3 Document Object Key
- Application Timestamp

## 6. Environment Configuration Reference (.env)

| Variable              | Required | Description                                  |
|-----------------------|----------|----------------------------------------------|
| AWS_ACCESS_KEY_ID     | Yes      | IAM Access Key ID                            |
| AWS_SECRET_ACCESS_KEY | Yes      | IAM Secret Access Key                        |
| AWS_REGION            | Yes      | AWS Region for S3, SES, etc. (eu-north-1)    |
| AWS_S3_BUCKET_NAME    | Yes      | Target S3 bucket for resumes                 |
| AWS_SESSION_TOKEN     | Optional | STS session token (Academy/Labs)             |
| AWS_SES_SENDER_EMAIL  | Optional | Verified sender email for SES                |
| RECRUITER_PASSCODE    | Optional | Recruiter dashboard password                 |
| PORT                  | Optional | Backend HTTP server port (5000)              |
| NODE_ENV              | Optional | Application runtime environment (production) |

## 7. Production Deployment & Setup — Screenshots

This section documents the end-to-end cloud setup and deployment of CloudATS on AWS, from resource provisioning through to the live, running application, captured directly from the project's development and deployment session.

### 7.1 Amazon S3 Bucket Creation

An S3 bucket (dscproj) was created in the eu-north-1 (Stockholm) region to store candidate resumes in a partitioned folder structure.

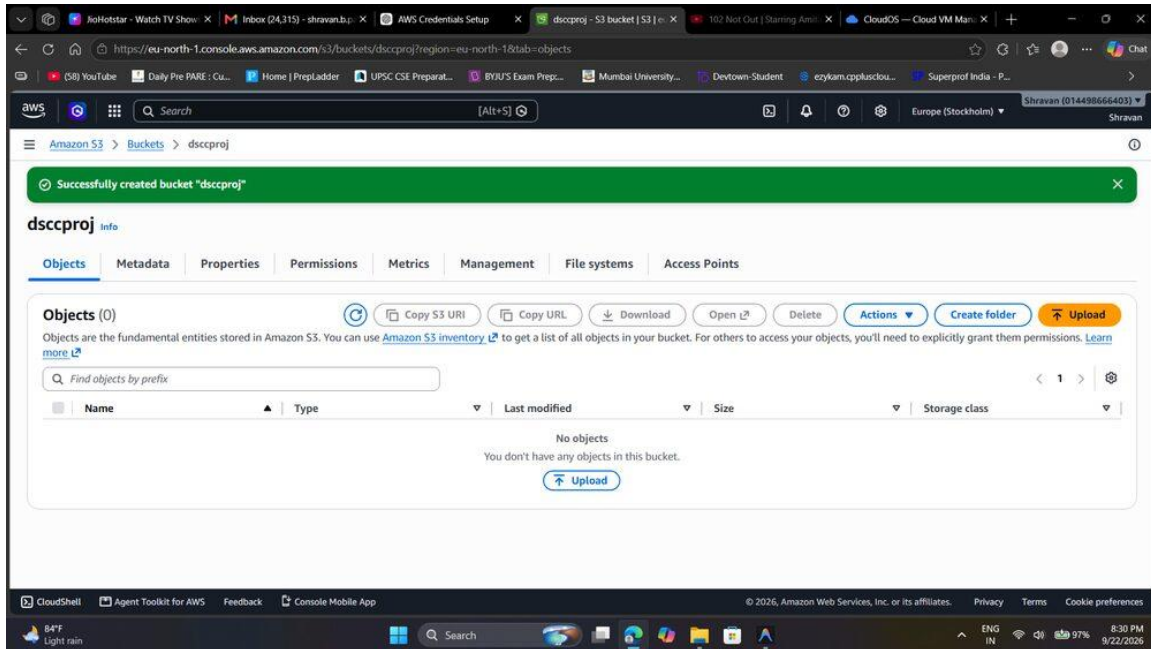


Fig 7.1 — S3 bucket "dscproj" successfully created on AWS.

### 7.2 IAM User, Policies & Access Keys Setup

A dedicated IAM user was created with a least-privilege custom policy scoped to S3, Textract, Rekognition, CloudWatch, and SES actions, and access keys were generated for programmatic access from the application.

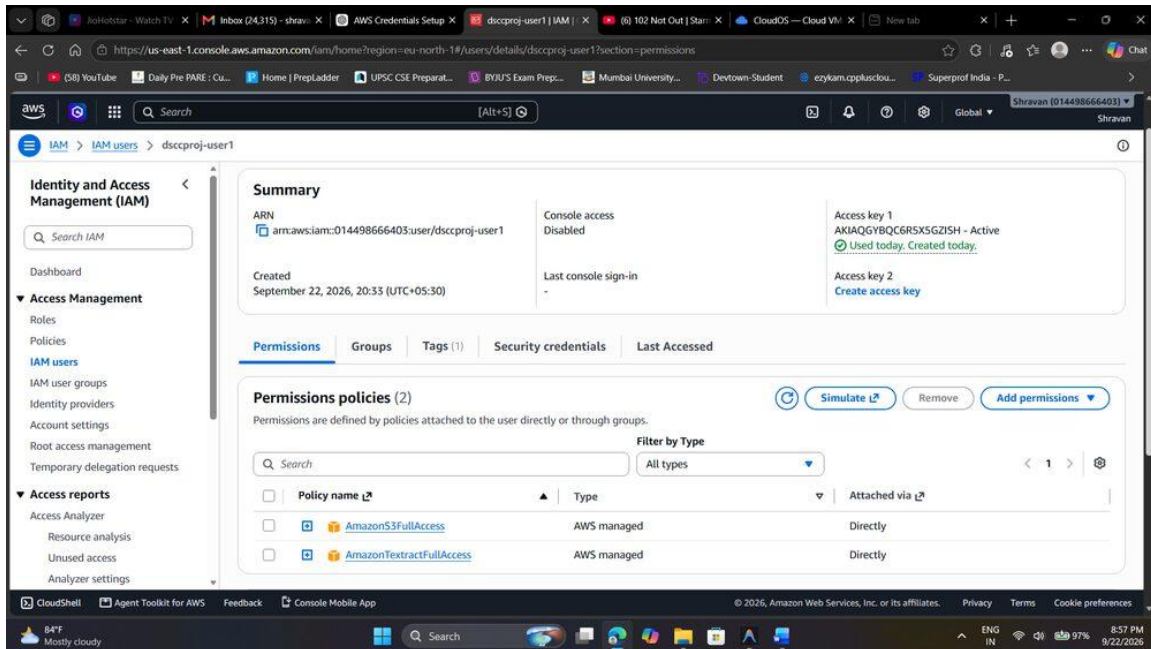


Fig 7.2a — IAM Identity and Access Management dashboard.

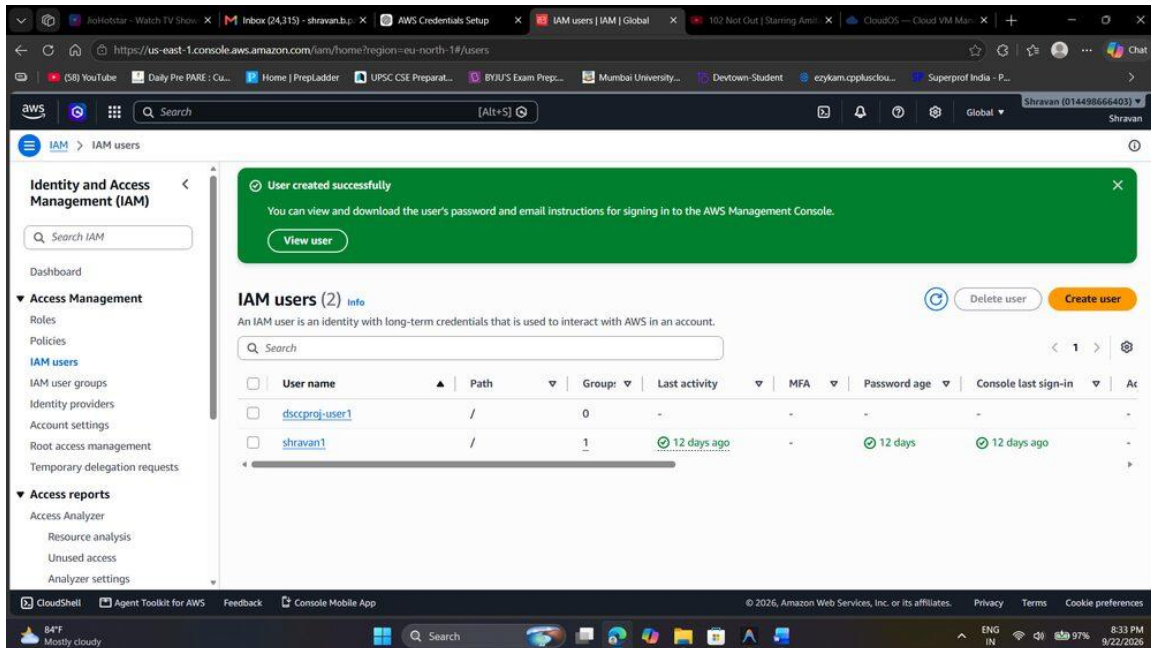


Fig 7.2b — IAM user summary and permissions policies.

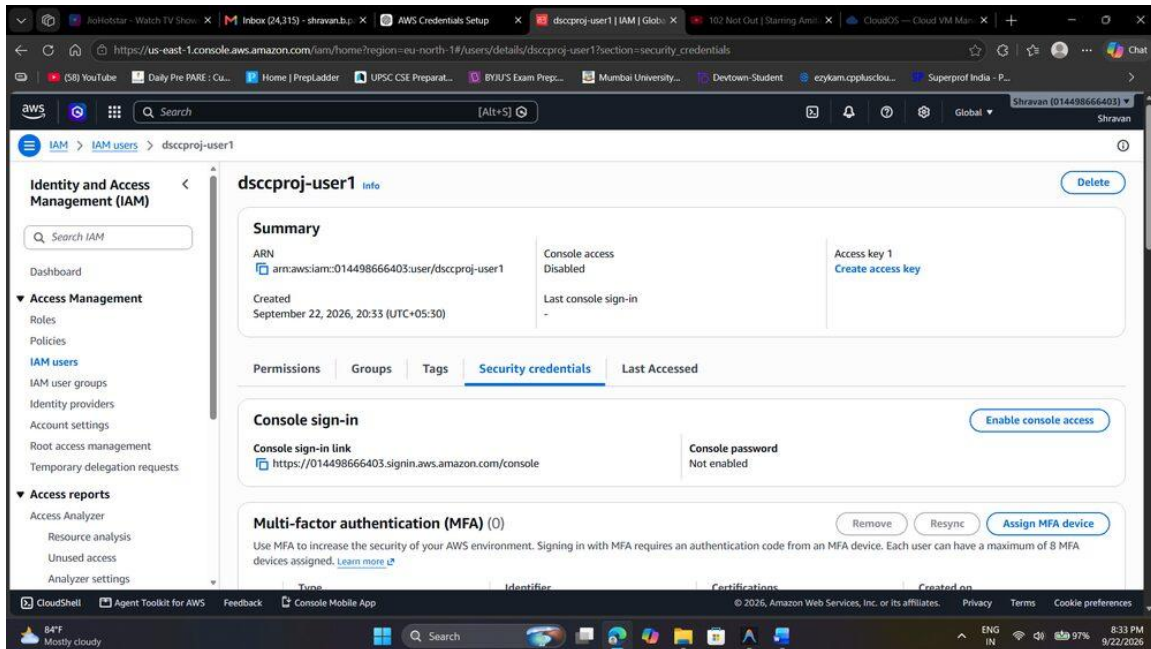


Fig 7.2c — IAM user configuration overview.

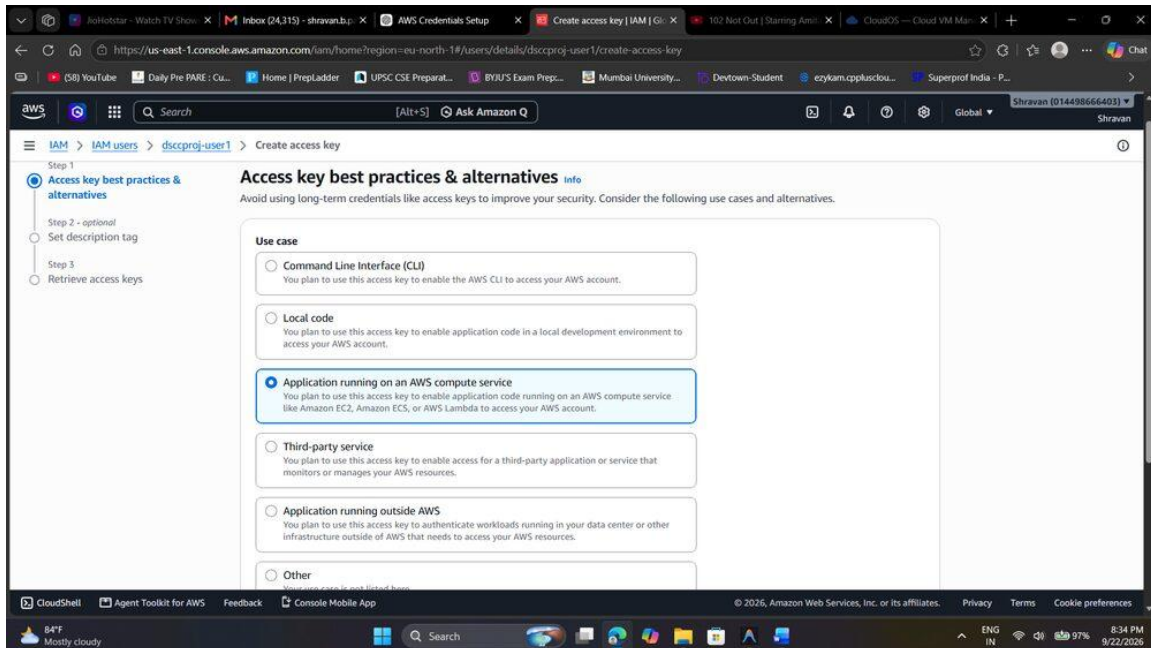


Fig 7.2d — Access key best practices and alternatives.

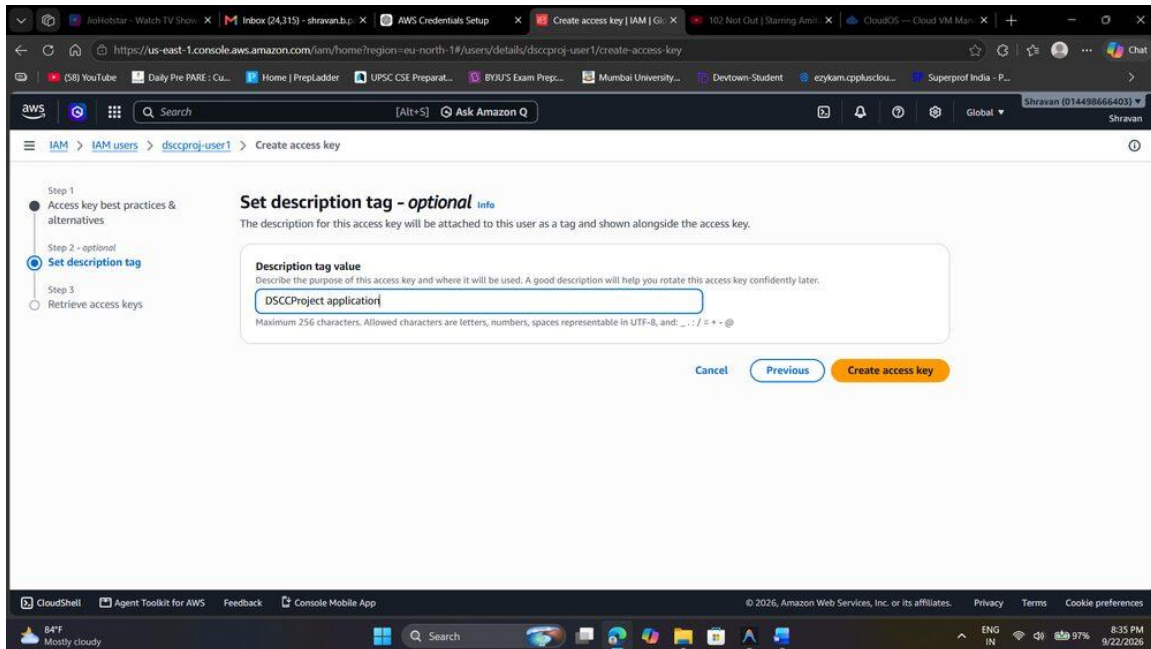


Fig 7.2e — Setting an optional description tag for the access key.

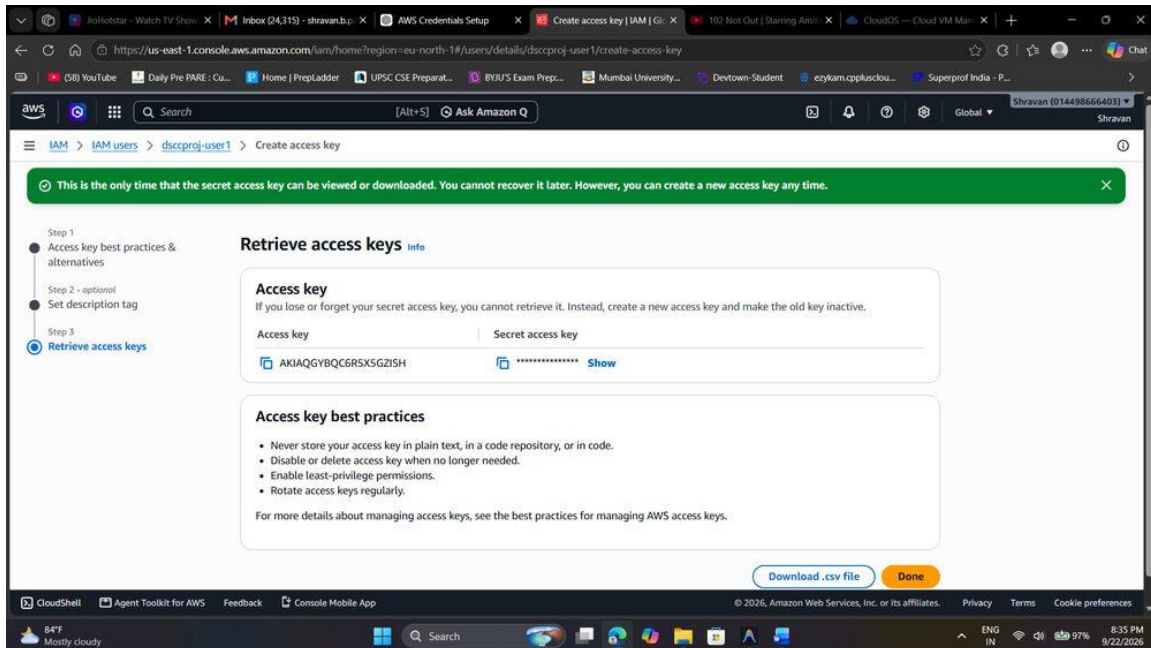


Fig 7.2f — Retrieved access key for the IAM user.

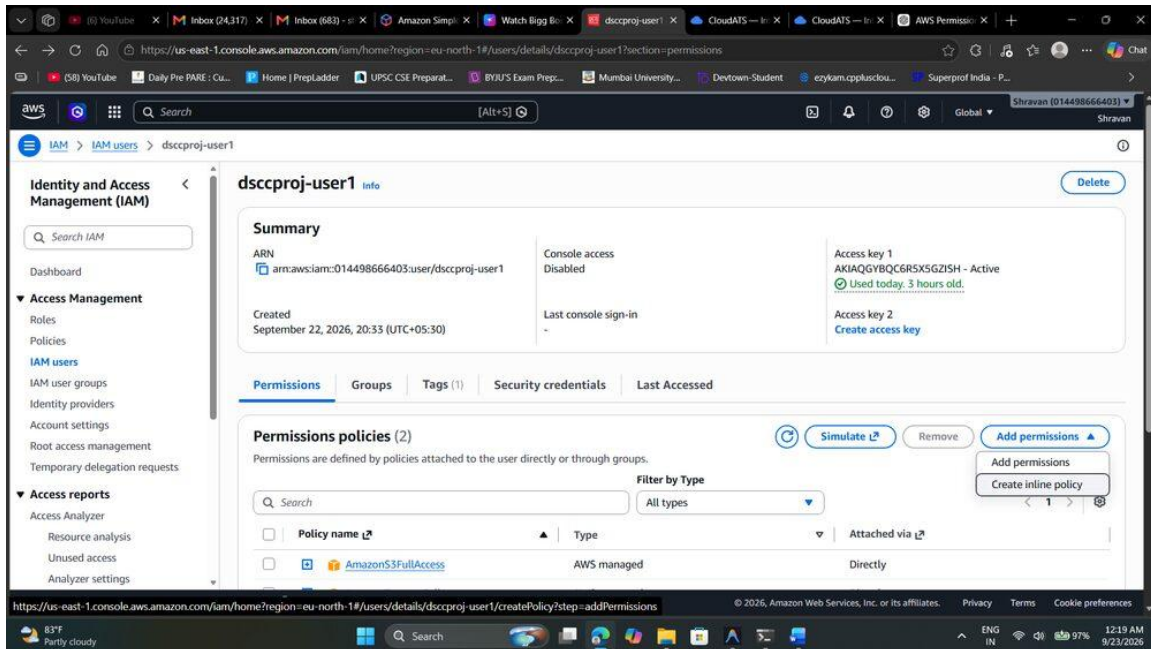


Fig 7.2g — dscproj-user1 IAM user summary and permissions policies.

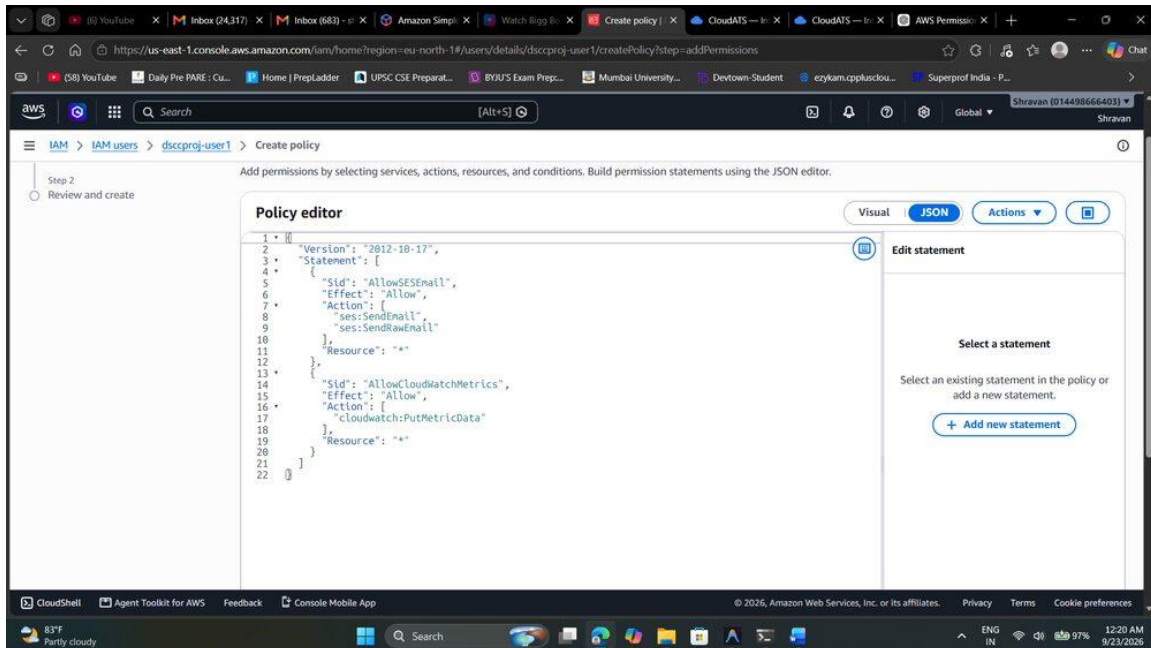


Fig 7.2h — Custom least-privilege policy editor (JSON).

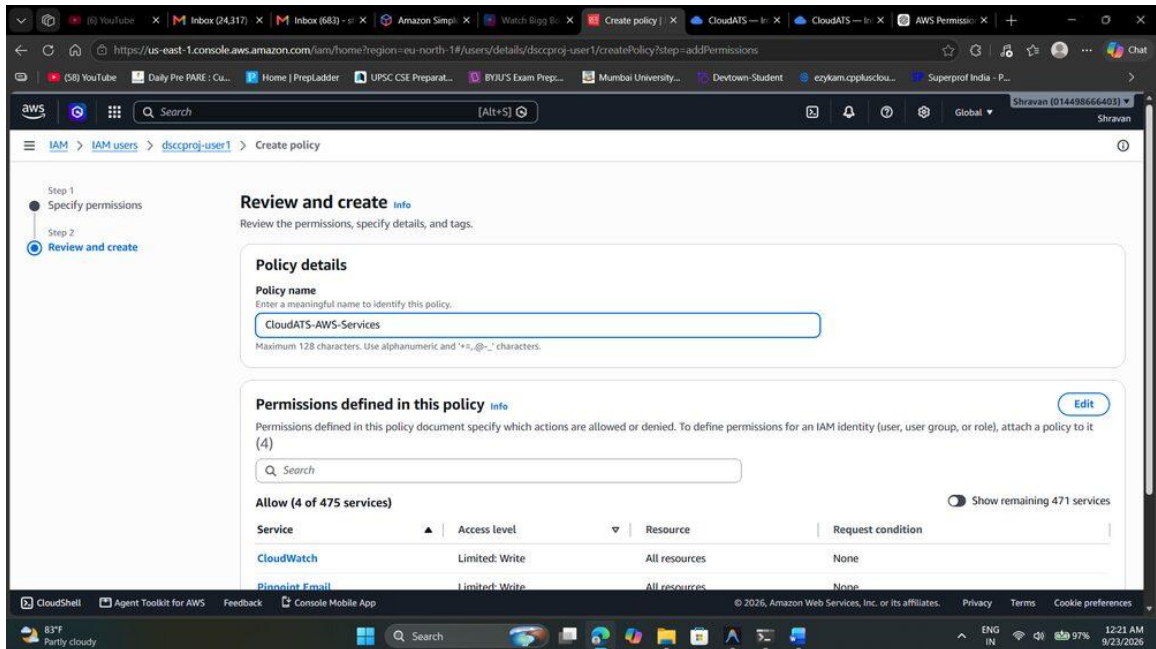


Fig 7.2i — Reviewing and creating the custom IAM policy.

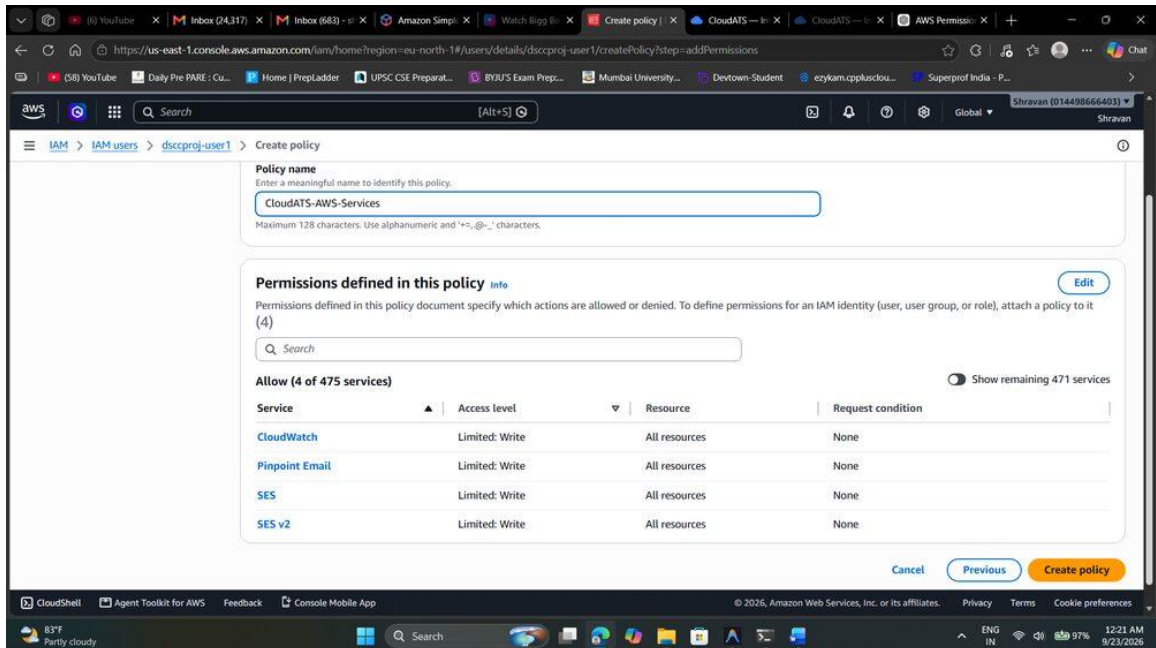


Fig 7.2j — Permissions defined in the custom policy.

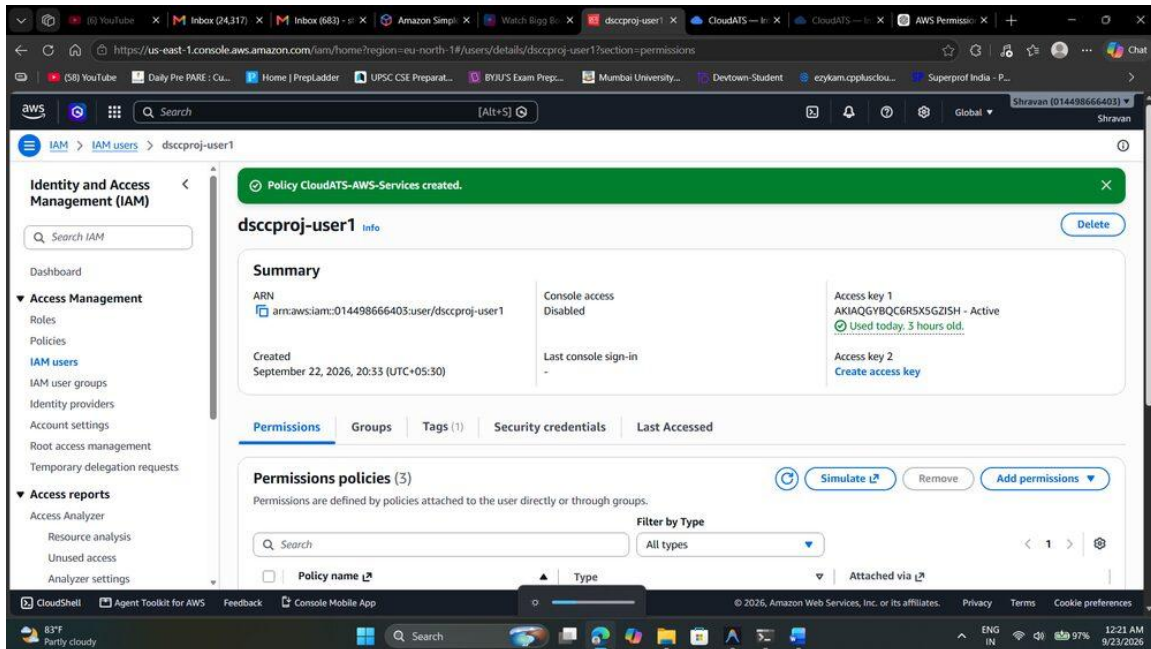


Fig 7.2k — Policy successfully attached to the IAM user (service enabled).

## 7.3 Amazon SES Setup & Email Identity Verification

Amazon SES was configured to send automated candidate application confirmation emails. A sender email identity was created and verified through AWS's email confirmation workflow.

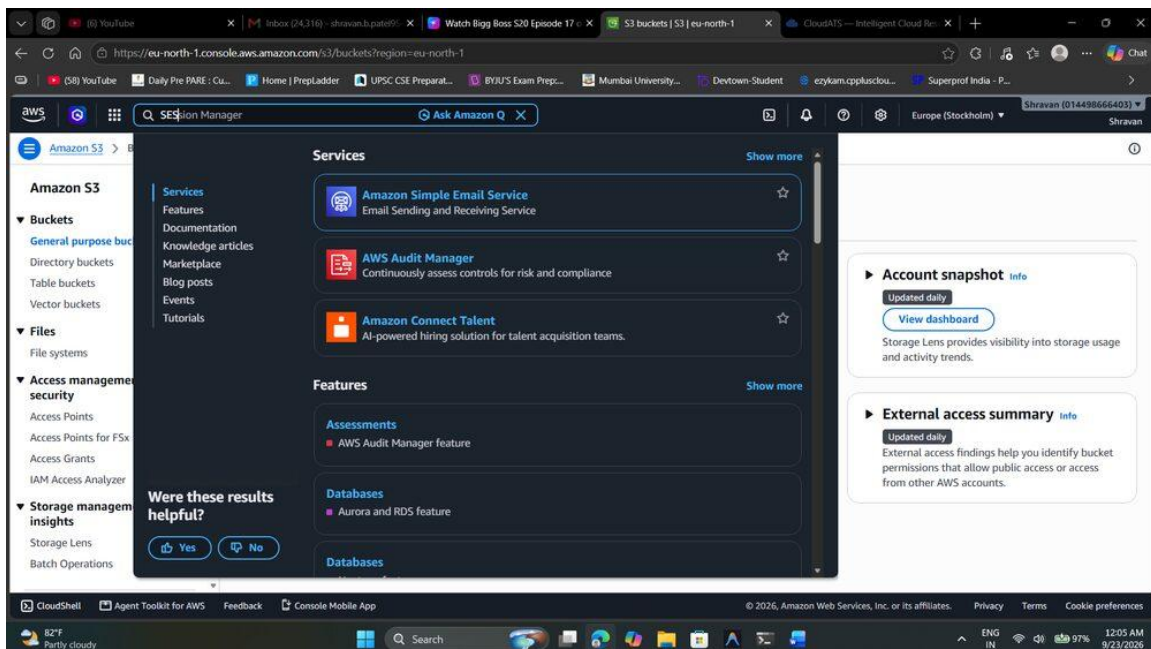


Fig 7.3a — AWS Services overview showing Amazon SES.

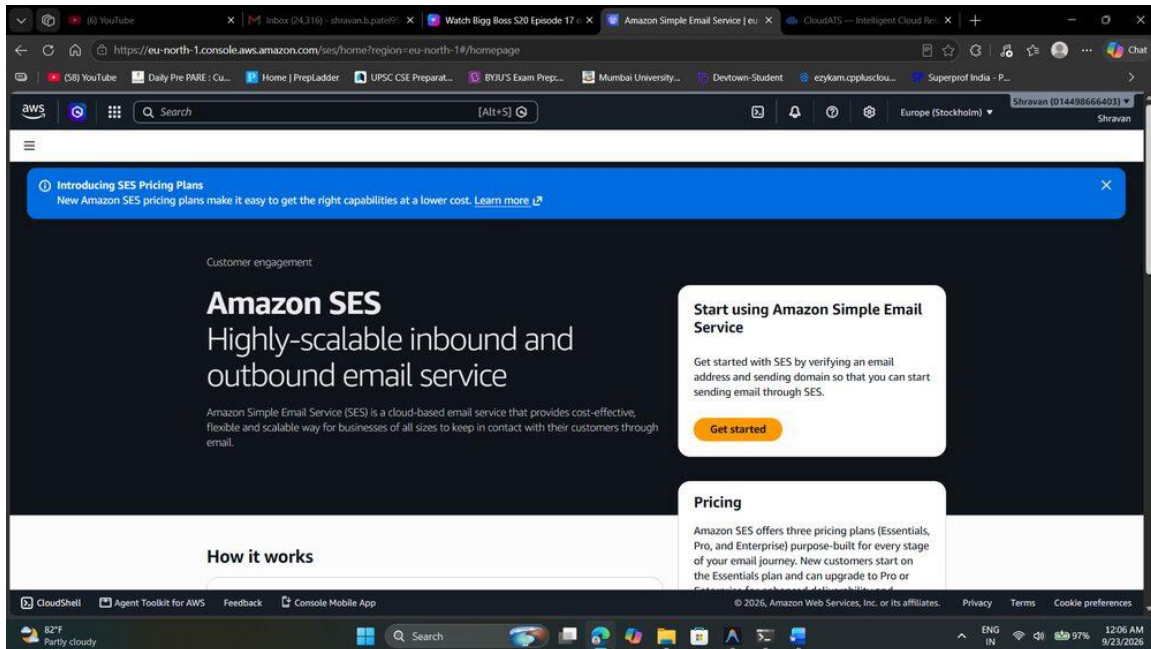


Fig 7.3b — Amazon SES service landing page.

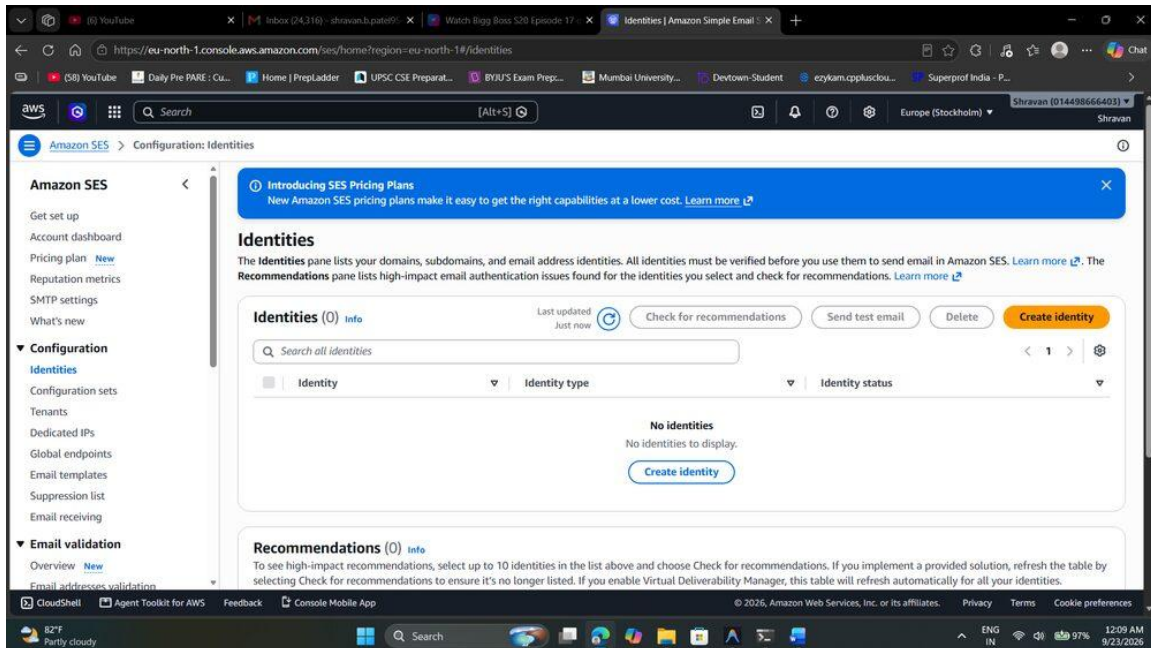


Fig 7.3c — Amazon SES dashboard and identity summary.

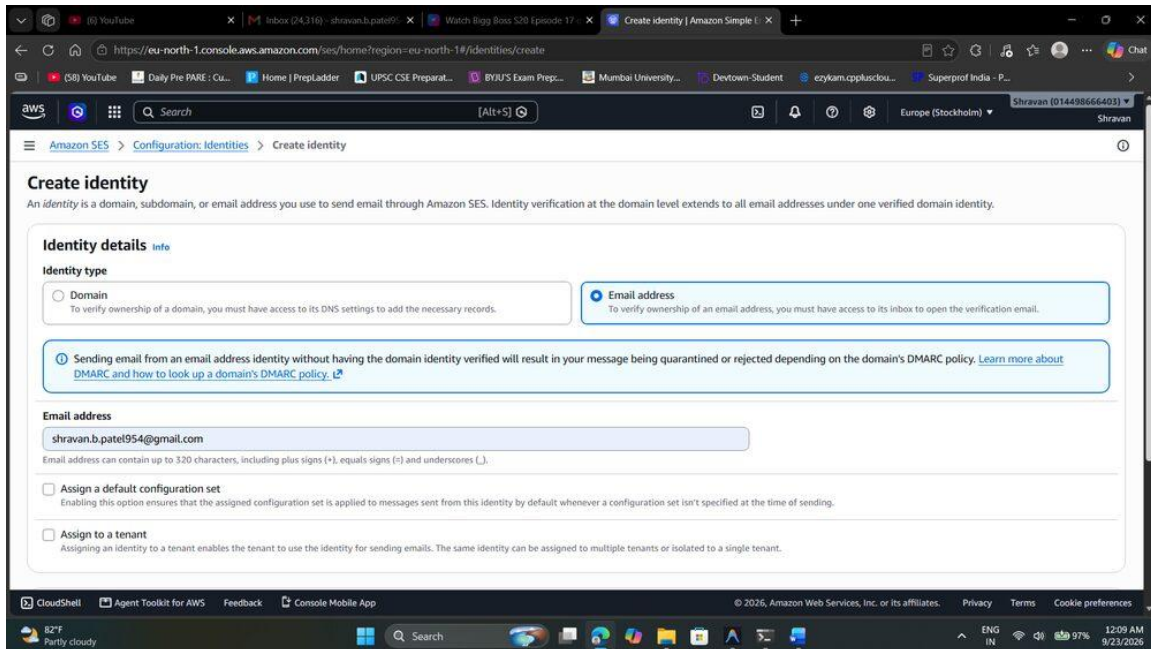


Fig 7.3d — Creating a new verified email identity in SES.

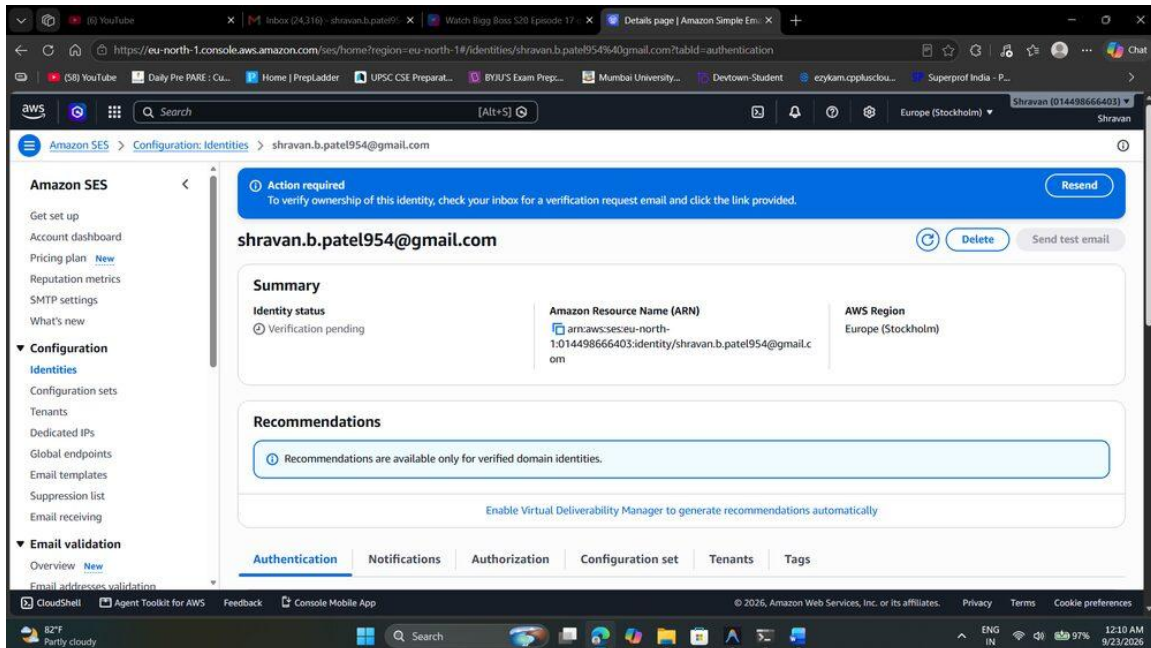


Fig 7.3e — SES verification email dispatched to the sender address.

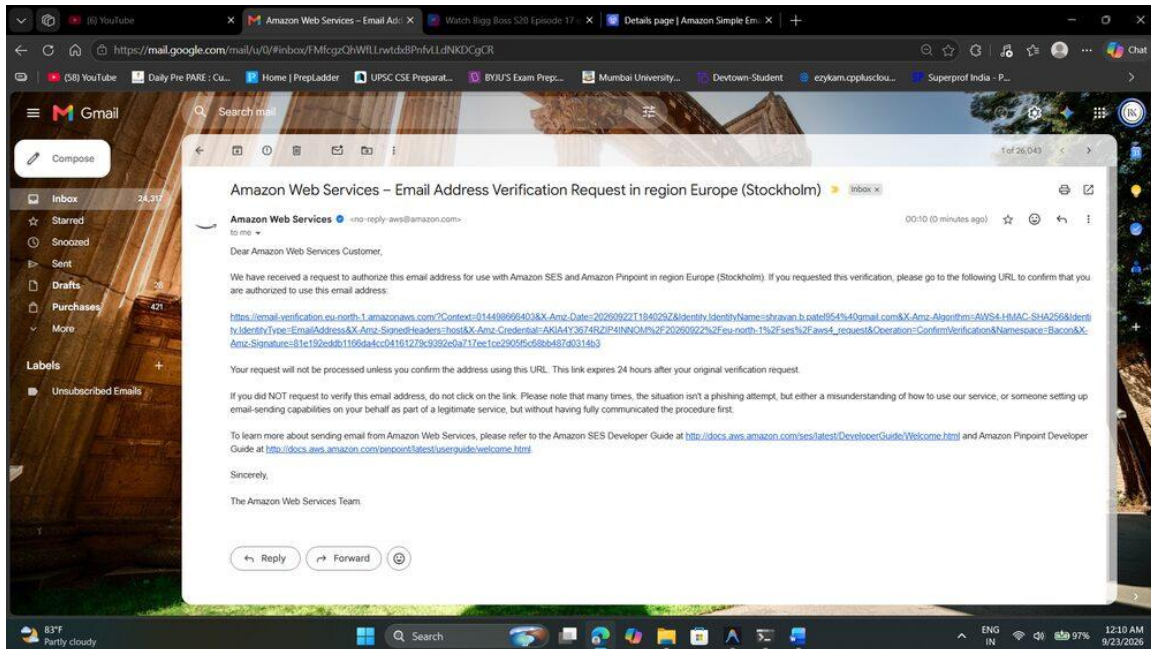


Fig 7.3f — AWS email address verification request received in inbox.

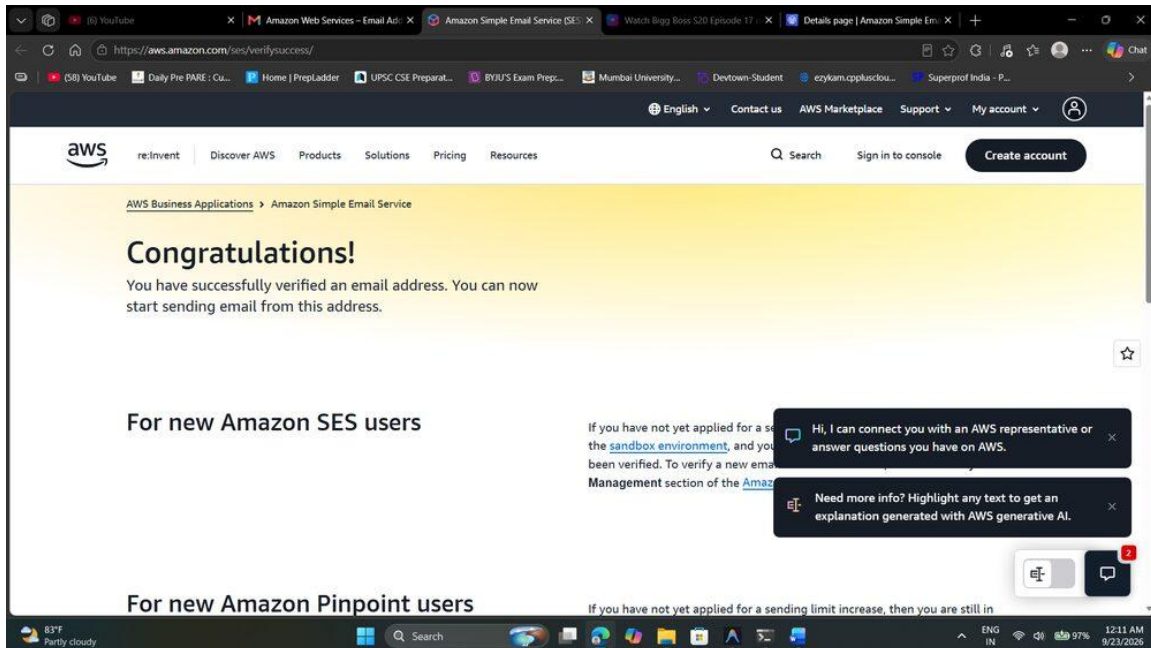


Fig 7.3g — Email address successfully verified confirmation.

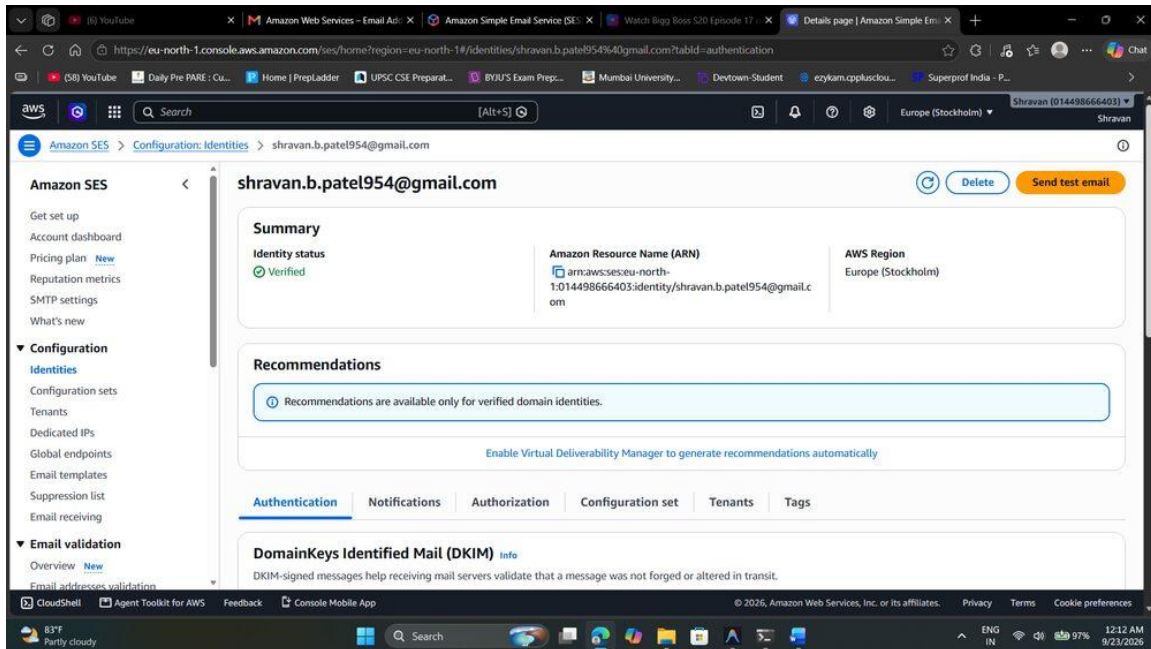


Fig 7.3h — Verified identity status shown in the SES console.

## 7.4 Amazon EC2 Instance Launch

An Amazon EC2 instance (Ubuntu 24.04 LTS) was provisioned to host the CloudATS backend and frontend.

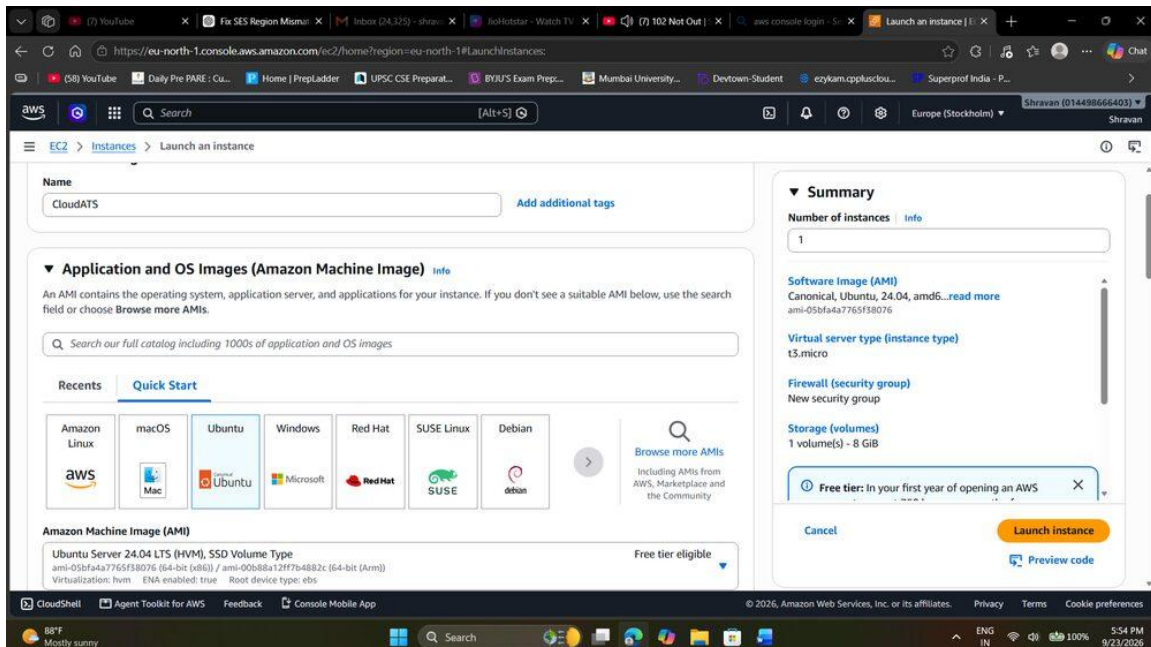


Fig 7.4a — Selecting the application/OS image (Ubuntu) for the instance.

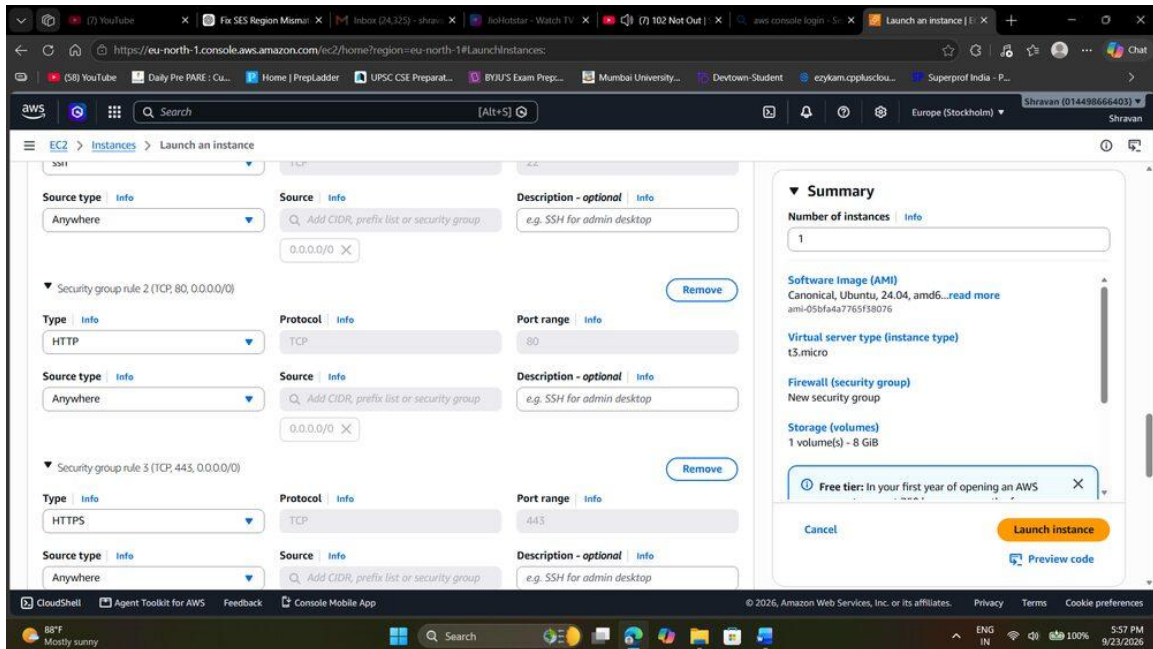


Fig 7.4b — Configuring instance type, key pair, and launch settings.

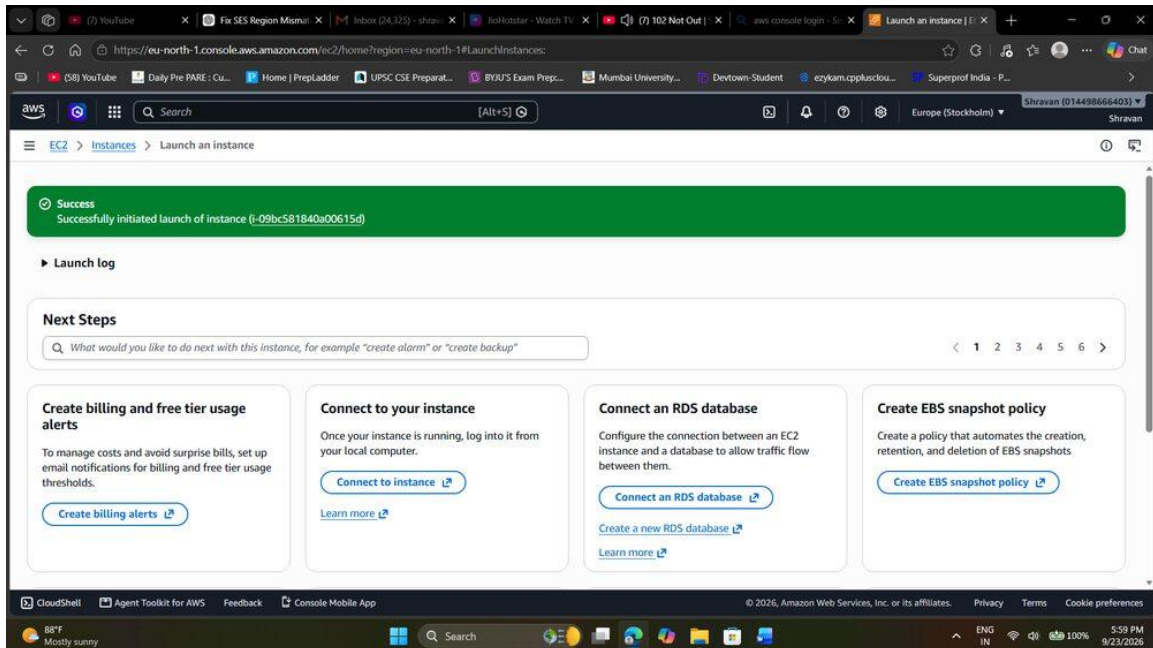


Fig 7.4c — EC2 instance successfully launched.

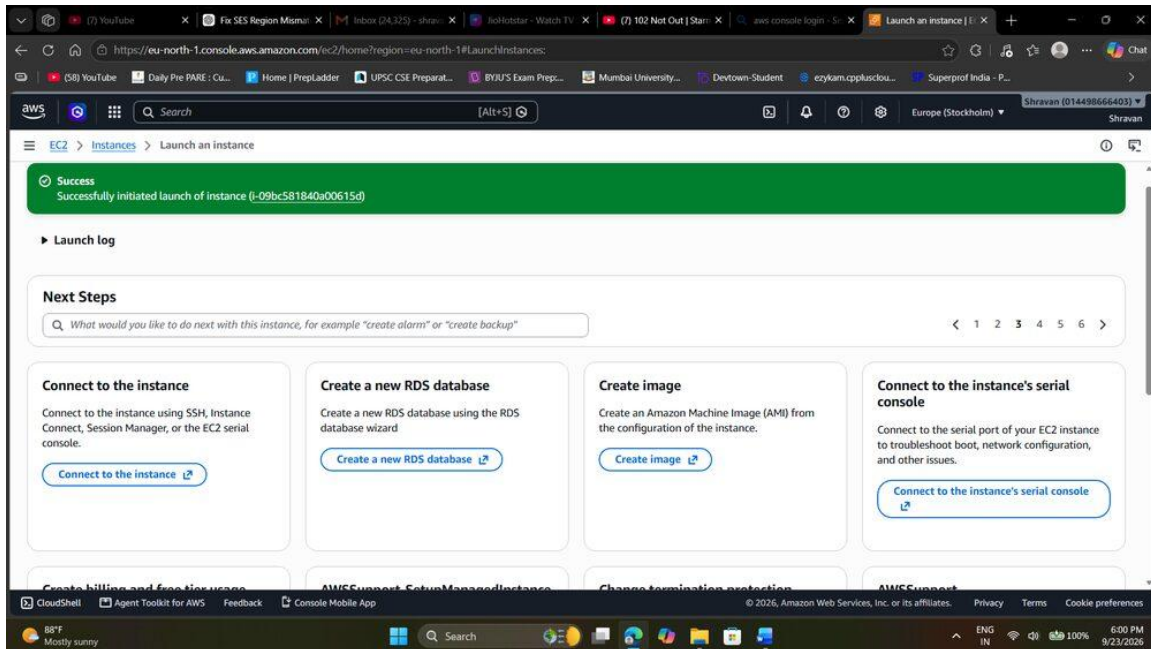


Fig 7.4d — Next steps after launch: connect, create alarms, and manage the instance.

## 7.5 Connecting to EC2 & Server Preparation

The instance was accessed over SSH, after which the OS packages, Git, and Node.js runtime were installed and verified.

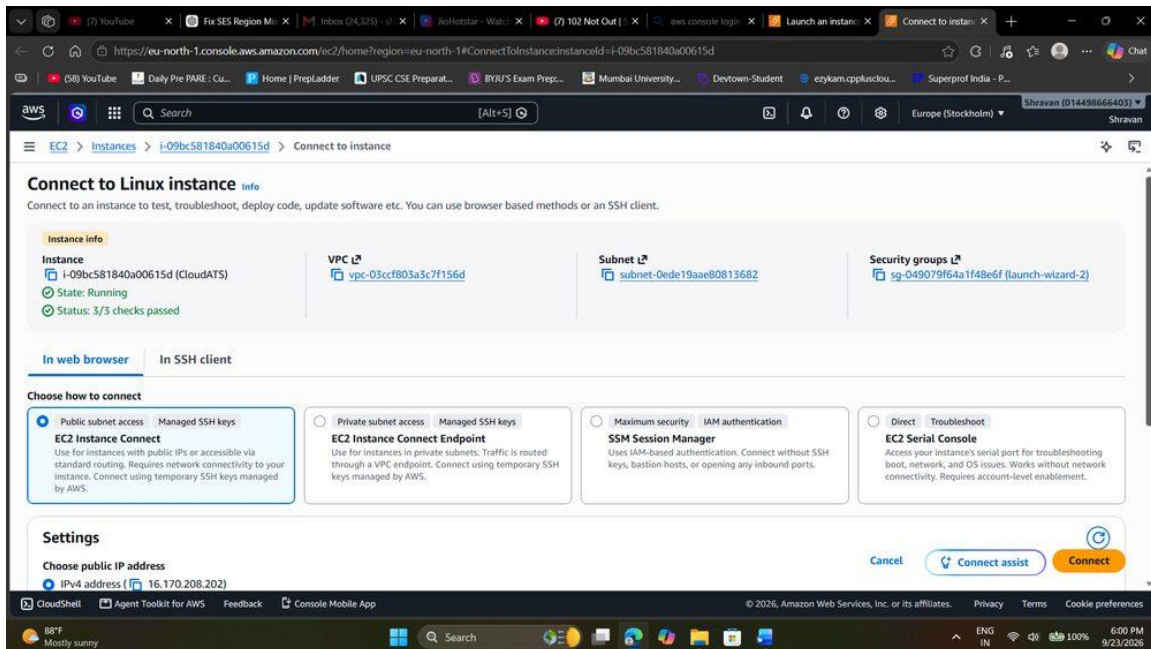


Fig 7.5a — Connect to Linux instance options (EC2 Instance Connect).

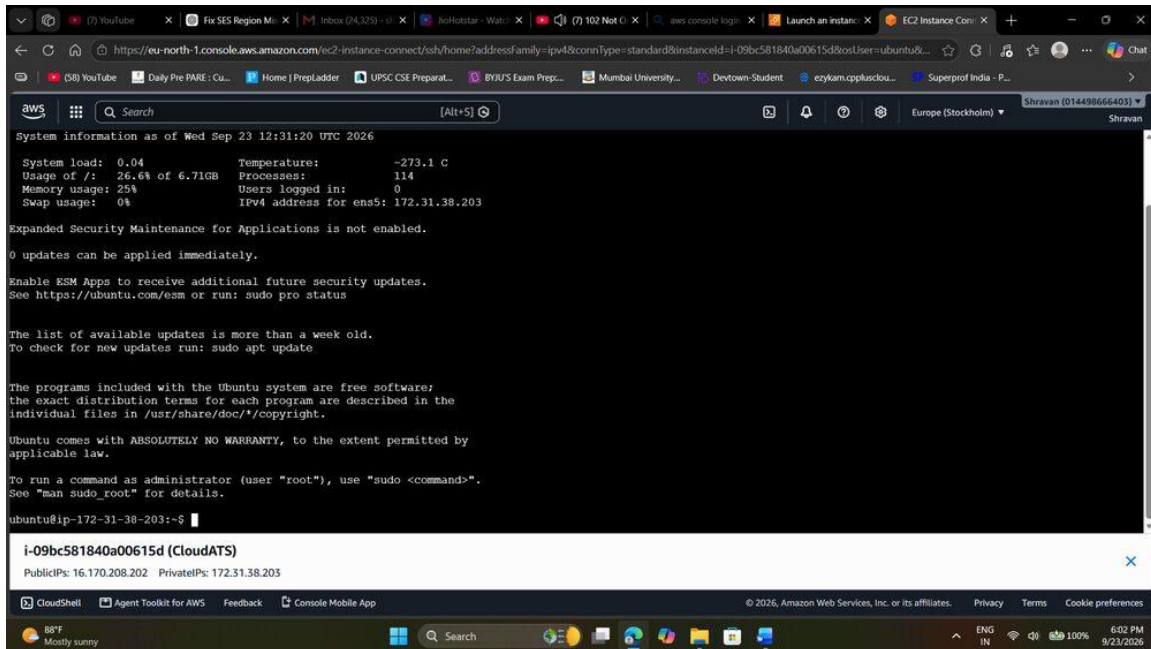


Fig 7.5b — Terminal session connected to the EC2 instance.

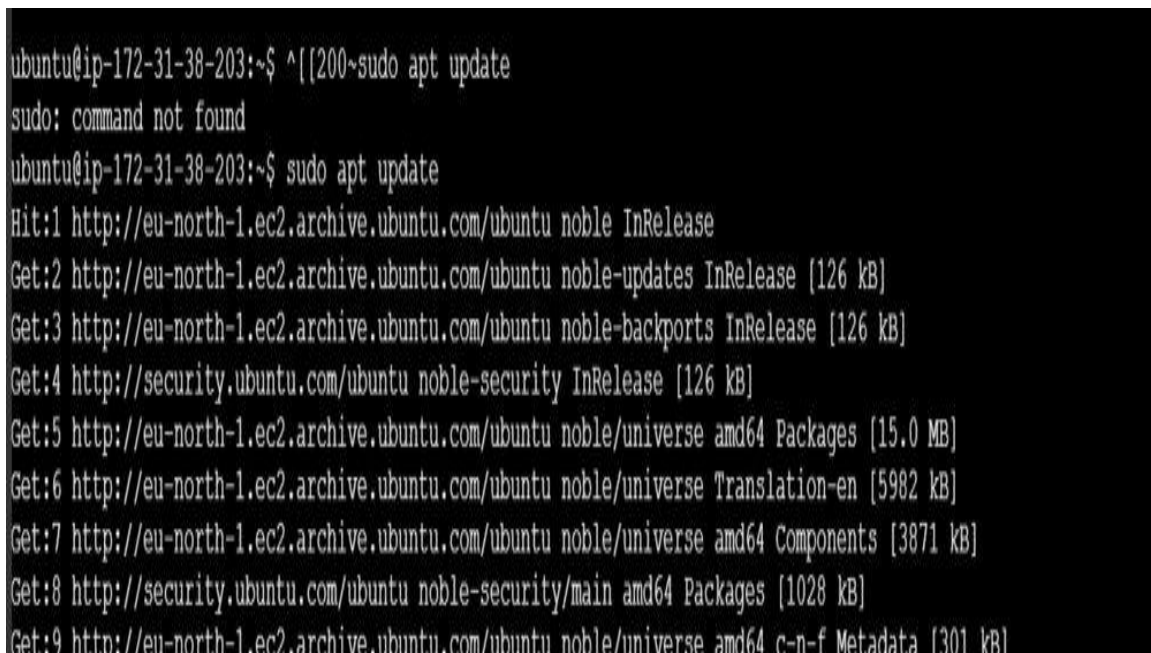


Fig 7.5c — Running `sudo apt update` on Ubuntu.

```

173 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-38-203:~$ sudo apt upgrade -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done

The following NEW packages will be installed:
  libdebuginfod-common libdebuginfod1t64 liblvm29 linux-aws-7.0-headers-7.0.0-1013 linux-aws-7.0-tools-7.0.0-1013 linux-headers-7.0.0-1013-aws
  linux-image-7.0.0-1013-aws linux-modules-7.0.0-1013-aws linux-tools-7.0.0-1013-aws

The following packages will be upgraded:
  amd64-microcode apport apport-core-dump-handler base-files bind9-dnsutils bind9-host bind9-libs bsdxtrautils bsduutils byobu ca-certificates console-setup
  console-setup-linux coreutils cpio curl diffutils dirmngr distro-info-data eject fdisk fwupd gawk gir1.2-glib-2.0 gnupg gnupg-l10n gnupg-utils gpg gpg-agent
  gpg-wks-client gpgconf gpgsm gpgv gzip iproute2 keyboard-configuration keyboxd kpartx krb5-locales libarchive13t64 libattr1 libblkid1 libbz2-1.0 libc-bin libc6
  libcurl3t64-gnutls libcurl4t64 libdrm-amdgpu1 libdrm-common libdrm2 libevent-core-2.1-7t64 libexpat1 libfdisk1 libfwupd3 libgcrypt20 libglib2.0-0t64 libglib2.0-bl

```

Fig 7.5d — apt update output and available package upgrades.

```

ubuntu@ip-172-31-38-203:~$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
git set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.

```

Fig 7.5e — Installing Git via `sudo apt install git`.

```
ubuntu@ip-172-31-38-203:~$ git --version
git version 2.43.0
ubuntu@ip-172-31-38-203:~$
```

Fig 7.5f — Verifying the Git installation with `git --version`.

```
ubuntu@ip-172-31-38-203:~$ curl -fsSL https://deb.nodesource.com/setup_22.x | sudo -E bash -
2026-09-23 12:38:13 - Installing pre-requisites
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ca-certificates is already the newest version (20260601~24.04.1).
ca-certificates set to manually installed.
curl is already the newest version (8.5.0-2ubuntu10.13).
curl set to manually installed.
gnupg is already the newest version (2.4.4-2ubuntu17.6).
```

Fig 7.5g — Running the NodeSource setup script for Node.js 22.x.

```
ubuntu@ip-172-31-38-203:~$ sudo apt install -y nodejs
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  nodejs
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 37.8 MB of archives.
After this operation, 236 MB of additional disk space will be used.
Get:1 https://deb.nodesource.com/node_22.x nodistro/main amd64 nodejs amd64 22.23.2-1nodesource1 [37.8 MB]
Fetched 37.8 MB in 1s (46.3 MB/s)
Selecting previously unselected package nodejs.
```

*Fig 7.5h — Verifying Node.js and npm versions (node -v, npm -v).*

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-38-203:~$ node -v
v22.23.2
ubuntu@ip-172-31-38-203:~$ npm -v
10.9.8
ubuntu@ip-172-31-38-203:~$
```

*Fig 7.5i — Confirming Node.js runtime readiness on the instance.*

## 7.6 Cloning the Repository & Building the Application

The CloudATS-CareerPortal repository was cloned from GitHub, dependencies were installed, and the production React frontend was built.

```

ubuntu@ip-172-31-38-203:~$ git clone https://github.com/shravanbpatel954/CloudATS-CareerPortal.git
Cloning into 'CloudATS-CareerPortal'...
Username for 'https://github.com': shravanbpatel954
Password for 'https://shravanbpatel954@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/shravanbpatel954/CloudATS-CareerPortal.git/'
ubuntu@ip-172-31-38-203:~$ git clone https://github.com/shravanbpatel954/CloudATS-CareerPortal.git
Cloning into 'CloudATS-CareerPortal'...
Username for 'https://github.com': shravanbpatel954
Password for 'https://shravanbpatel954@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/shravanbpatel954/CloudATS-CareerPortal.git/'
ubuntu@ip-172-31-38-203:~$ git clone https://github.com/shravanbpatel954/CloudATS-CareerPortal.git
Cloning into 'CloudATS-CareerPortal'...
Username for 'https://github.com': shravanbpatel954
Password for 'https://shravanbpatel954@github.com':
remote: Write access to repository not granted.
fatal: unable to access 'https://github.com/shravanbpatel954/CloudATS-CareerPortal.git/': The requested URL returned error: 403
ubuntu@ip-172-31-38-203:~$ git clone https://github.com/shravanbpatel954/CloudATS-CareerPortal.git
Cloning into 'CloudATS-CareerPortal'...
Username for 'https://github.com': shravanbpatel954
Password for 'https://shravanbpatel954@github.com':
remote: Enumerating objects: 37, done.
remote: Counting objects: 100% (37/37), done.
remote: Compressing objects: 100% (35/35), done.
remote: Total 37 (delta 4), reused 28 (delta 1), pack-reused 0 (from 0)
Receiving objects: 100% (37/37), 7.18 MiB | 13.08 MiB/s, done.
Resolving deltas: 100% (4/4), done.
ubuntu@ip-172-31-38-203:~$

```

Fig 7.6a — Cloning the repository with git clone and entering the project directory.

```

ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$ npm install

added 360 packages, and audited 361 packages in 20s

70 packages are looking for funding
  run `npm fund` for details

4 vulnerabilities (3 moderate, 1 high)

To address all issues (including breaking changes), run:
  npm audit fix --force

Run `npm audit` for details.
npm notice
npm notice New major version of npm available! 10.9.8 -> 12.1.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v12.1.0
npm notice To update run: npm install -g npm@12.1.0
npm notice
ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$

```

Fig 7.6b — Running npm install to install project dependencies.

```
ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$ npm run build

> cloudos-frontend@1.0.0 build
> vite build

vite v5.4.21 building for production...
✓ 1651 modules transformed.
dist/index.html                1.37 kB | gzip: 0.78 kB
dist/assets/index-Dv3eQHFe.css 29.72 kB | gzip: 5.85 kB
dist/assets/index-B6JjEtka.js 262.01 kB | gzip: 79.40 kB
✓ built in 5.30s

ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$
```

*Fig 7.6c — Running npm run build to generate the production frontend bundle.*

## 7.7 Environment Configuration (.env)

The environment configuration file was created on the server and populated with AWS credentials, region, bucket name, and application settings.

```
GNU nano 7.2 .env *
AWS_ACCESS_KEY_ID=AKIAQGYBQC6R5X5GZISH
AWS_SECRET_ACCESS_KEY=mVDo3rykm8YphlM4znCeN6Theim/LtmmMlIs9j5
AWS_REGION=eu-north-1
AWS_SQS_SENDER_EMAIL=shravan.b.patel1954@gmail.com
AWS_S3_BUCKET_NAME=dscproj
AWS_SESSION_TOKEN=
RECRUITER_PASSCODE=cloudats2026[]
```

*Fig 7.7a — Creating the .env file using the nano editor.*



Fig 7.7b — Editing environment variables in the .env file.

## 7.8 PM2 Process Management & Application Start

PM2 was installed globally to keep the Node.js server running continuously in the background, with automatic restarts on crash or reboot.



Fig 7.8a — Installing PM2 globally via npm.

[illegible]

```
[PM2] Spawning PM2 daemon with pm2_home=/home/ubuntu/.pm2
[PM2] PM2 Successfully daemonized
[PM2] Starting /home/ubuntu/CloudATS-CareerPortal/server/server.js in fork_mode (1 instance)
[PM2] Done.
```

| id | name     | namespace | version | mode | pid   | uptime | 0 | status | cpu | mem    | user   | watching |
|----|----------|-----------|---------|------|-------|--------|---|--------|-----|--------|--------|----------|
| 0  | cloudats | default   | 1.0.0   | fork | 28113 | 0s     | 0 | online | 0%  | 29.8mb | ubuntu | disabled |

```
ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$
```

*Fig 7.8c — Starting the CloudATS server with pm2 start and viewing pm2 status.*

[illegible]

*Fig 7.8d — Running pm2 save to persist the process list.*

```
ExecStart=/usr/lib/node_modules/pm2/bin/pm2 resurrect
ExecReload=/usr/lib/node_modules/pm2/bin/pm2 reload all
ExecStop=/usr/lib/node_modules/pm2/bin/pm2 kill

[Install]
WantedBy=multi-user.target

Target path
/etc/systemd/system/pm2-ubuntu.service

Command list
[ 'systemctl enable pm2-ubuntu' ]

[PM2] Writing init configuration in /etc/systemd/system/pm2-ubuntu.service
[PM2] Making script booting at startup...
[PM2] [-] Executing: systemctl enable pm2-ubuntu...
Created symlink /etc/systemd/system/multi-user.target.wants/pm2-ubuntu.service -> /etc/systemd/system/pm2-ubuntu.service.
[PM2] [v] Command successfully executed.
+-----+
[PM2] Freeze a process list on reboot via:
$ pm2 save

[PM2] Remove init script via:
$ pm2 unstartup systemd
ubuntu@ip-172-31-38-203:~/CloudATS-CareerPortal$
```

*Fig 7.8e — Configuring pm2 startup for automatic restart on reboot.*

## 7.9 AWS Security Group Configuration

The EC2 instance's security group was updated to allow inbound traffic on port 5000, making the application publicly reachable.

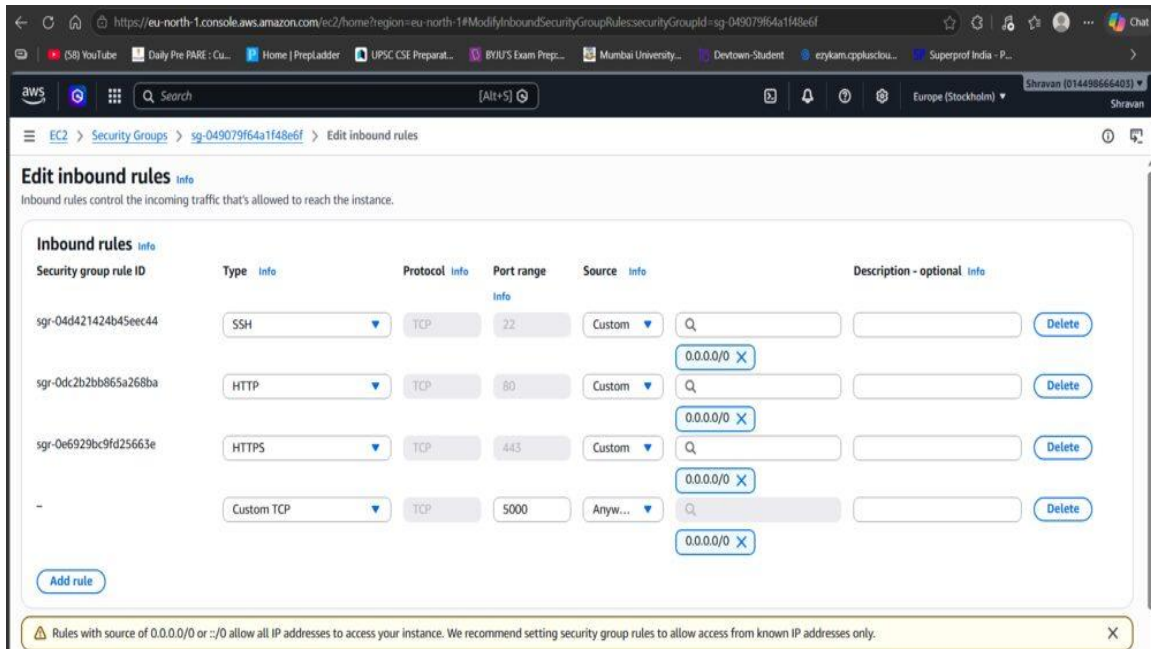


Fig 7.9a — Editing inbound rules to open custom TCP port 5000.

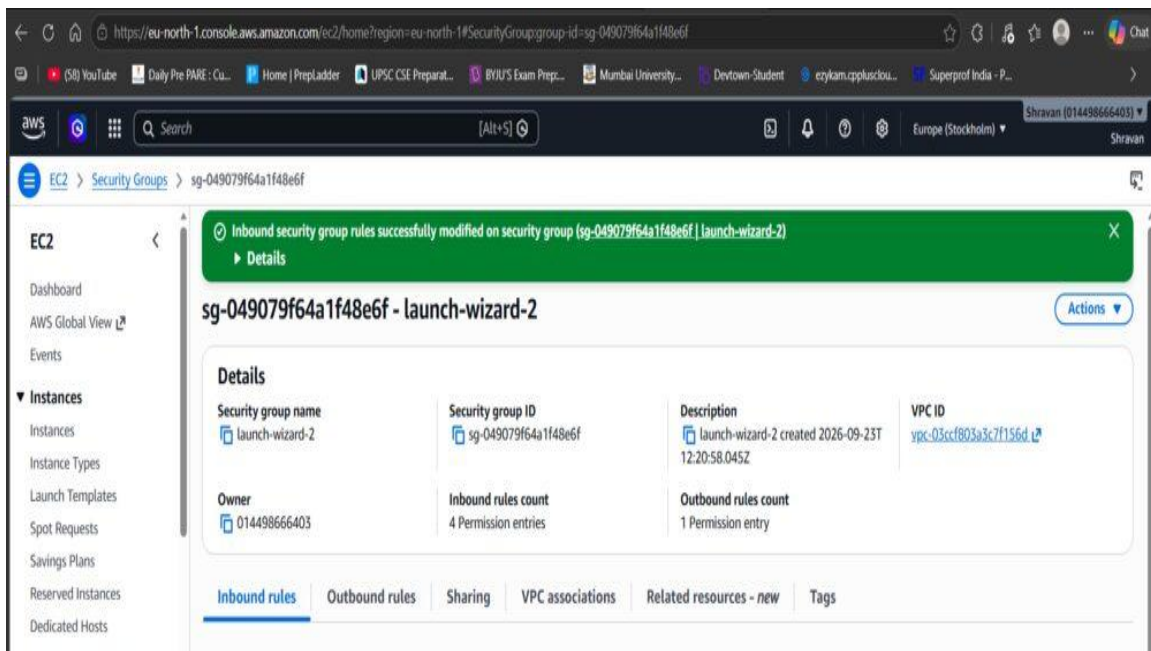


Fig 7.9b — Security group rules successfully updated.

## 7.10 Live Application — Candidate Portal Walkthrough

With the server running and reachable, the following screenshots demonstrate the live candidate-facing application flow, from login to application submission and confirmation.

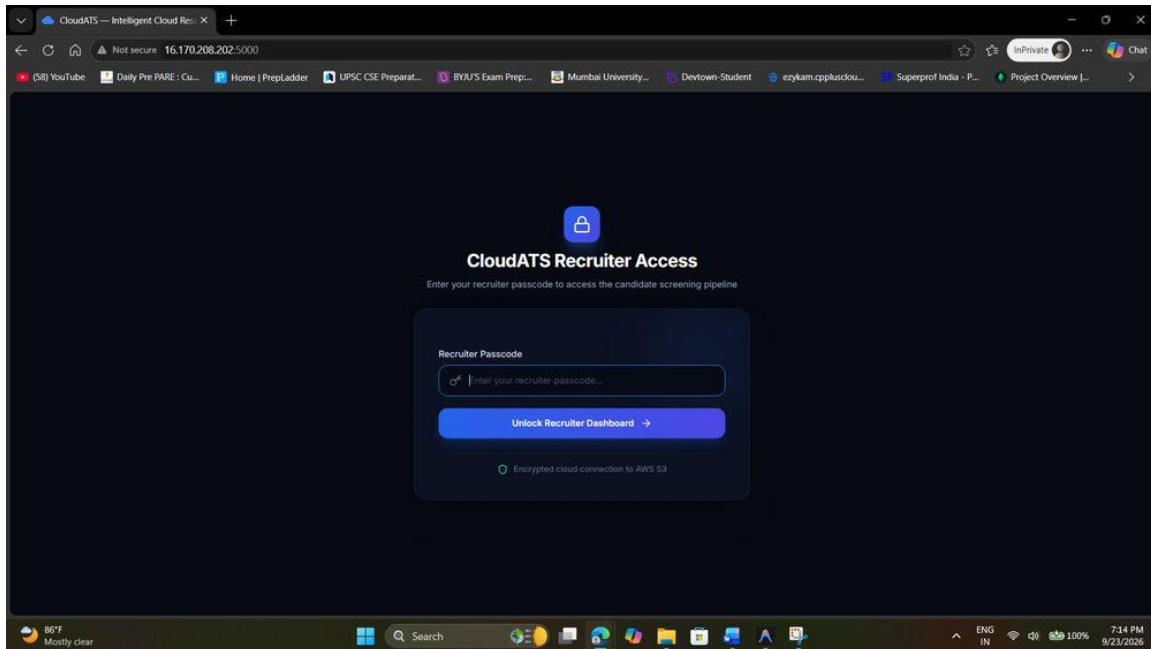


Fig 7.10a — CloudATS Recruiter Access login screen.

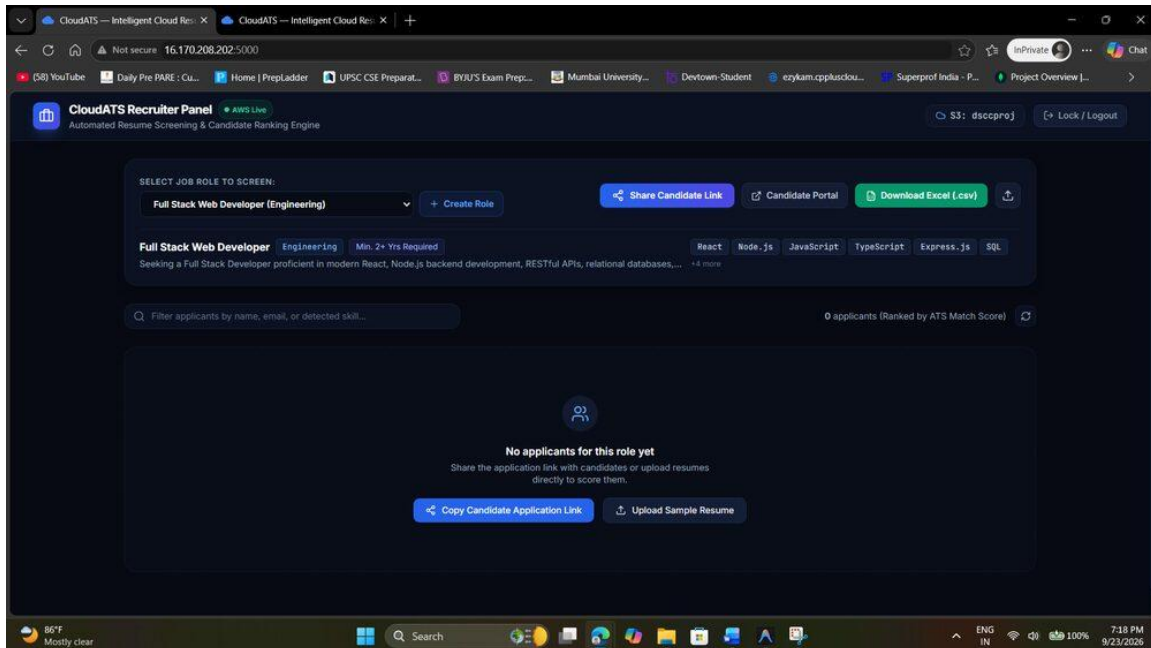


Fig 7.10b — Candidate application form and job listing.

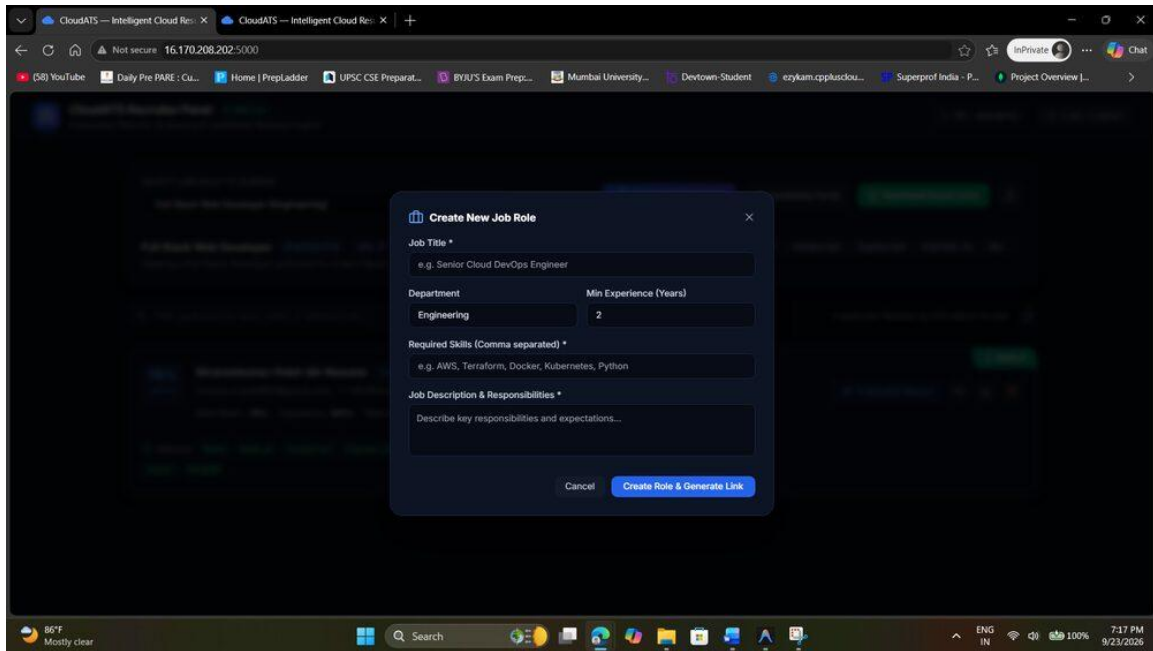


Fig 7.10c — Candidate submitting identity and profile details.

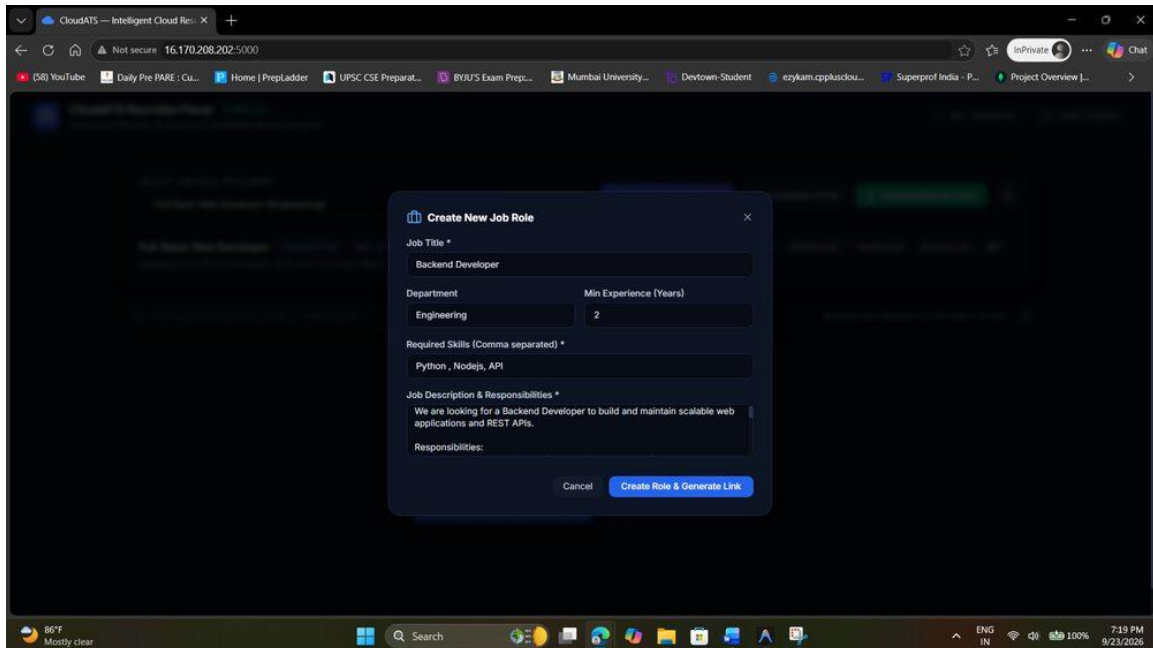


Fig 7.10d — Creating candidate profile and resume upload step.

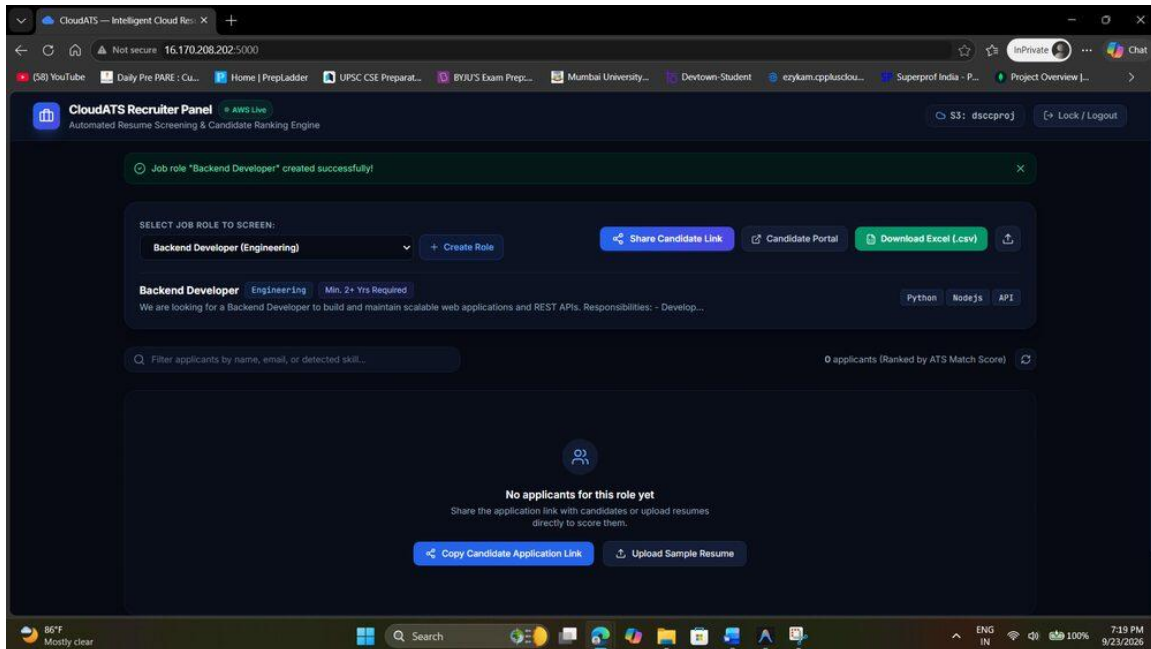


Fig 7.10e — Recruiter form dashboard after a job is posted.

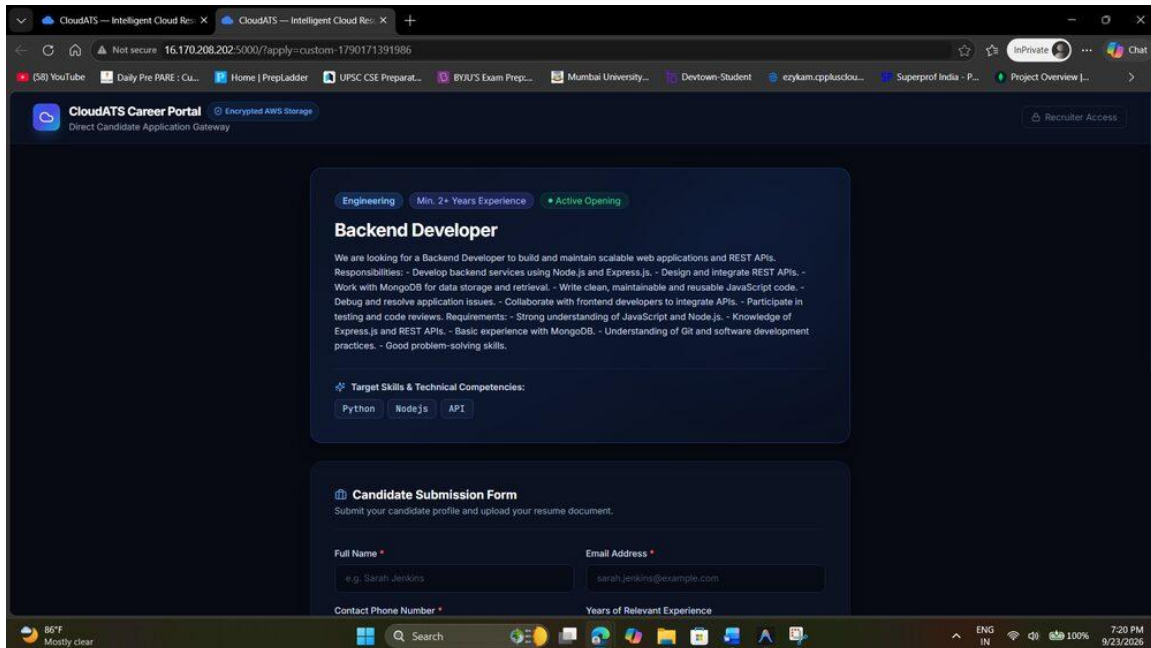


Fig 7.10f — Candidate-facing job listing page (Backend Developer role).

CloudATS Career Portal  
Direct Candidate Application Gateway

Submit your candidate profile and upload your resume document.

Full Name \*  
Shravankumar Patel

Email Address \*  
shravan.b.patel80@gmail.com

Contact Phone Number \*  
+918104479942

Years of Relevant Experience  
2

LinkedIn Profile or Portfolio URL  
https://shravankumar.co.in/

Cover Note / Key Highlights (Optional)  
Briefly highlight your experience with the required skills...

Upload Resume Document (.pdf, .docx, .doc, .txt) \*

Shravankumar\_Patel\_AICines.docx  
11 KB • Ready for AWS Ingestion  
Click to select different file

Submit Application

Fig 7.10g — Candidate application form with resume upload.

CloudATS Career Portal  
Direct Candidate Application Gateway

Debug and resolve application issues. - Collaborate with frontend developers to integrate APIs. - Participate in testing and code reviews. Requirements: - Strong understanding of JavaScript and Node.js. - Knowledge of Express.js and REST APIs. - Basic experience with MongoDB. - Understanding of Git and software development practices. - Good problem-solving skills.

Target Skills & Technical Competencies:  
Python Node.js API

Application Received & Stored in Cloud!  
Your resume has been uploaded directly to the AWS S3 recruitment vault and analyzed by our automated screening engine.

Application ID: app-1798171488716-55x79  
Target Role: Backend Developer  
AWS Cloud Status: ✓ Stored in S3

Submit Another Application

Fig 7.10h — "Application Received & Stored in Cloud" confirmation screen.

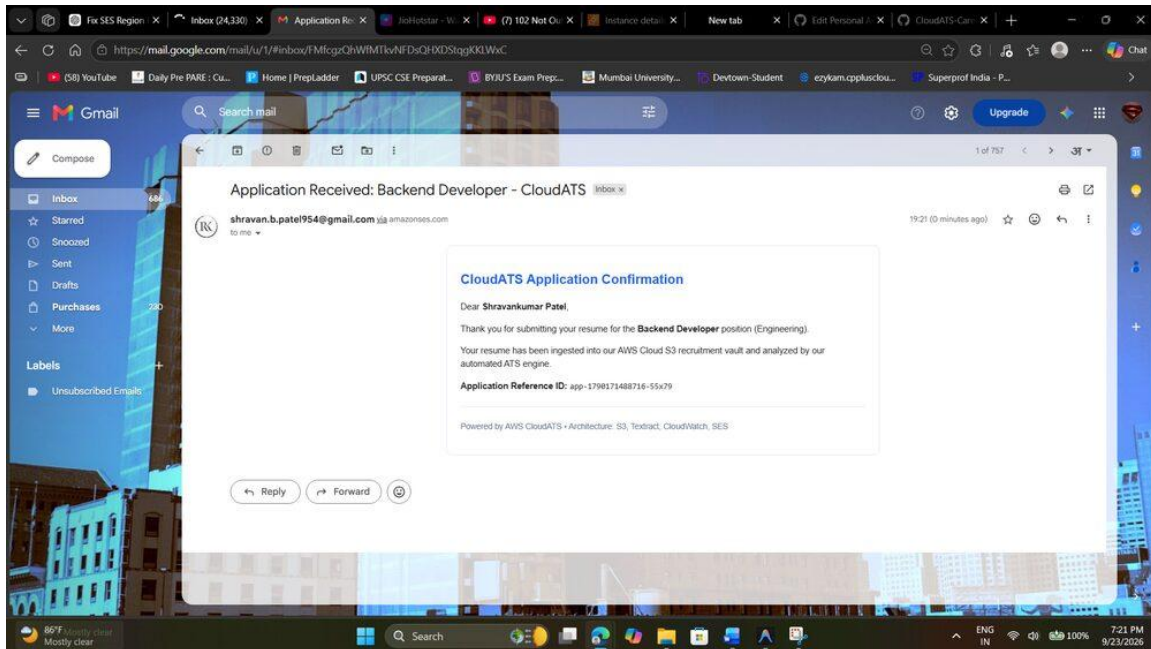


Fig 7.10i — Automated SES confirmation email received by the candidate.

## 7.11 Live Application — Recruiter Dashboard & Pipeline

The recruiter-facing dashboard provides ATS scoring, candidate ranking, evaluation details, and one-click data export.

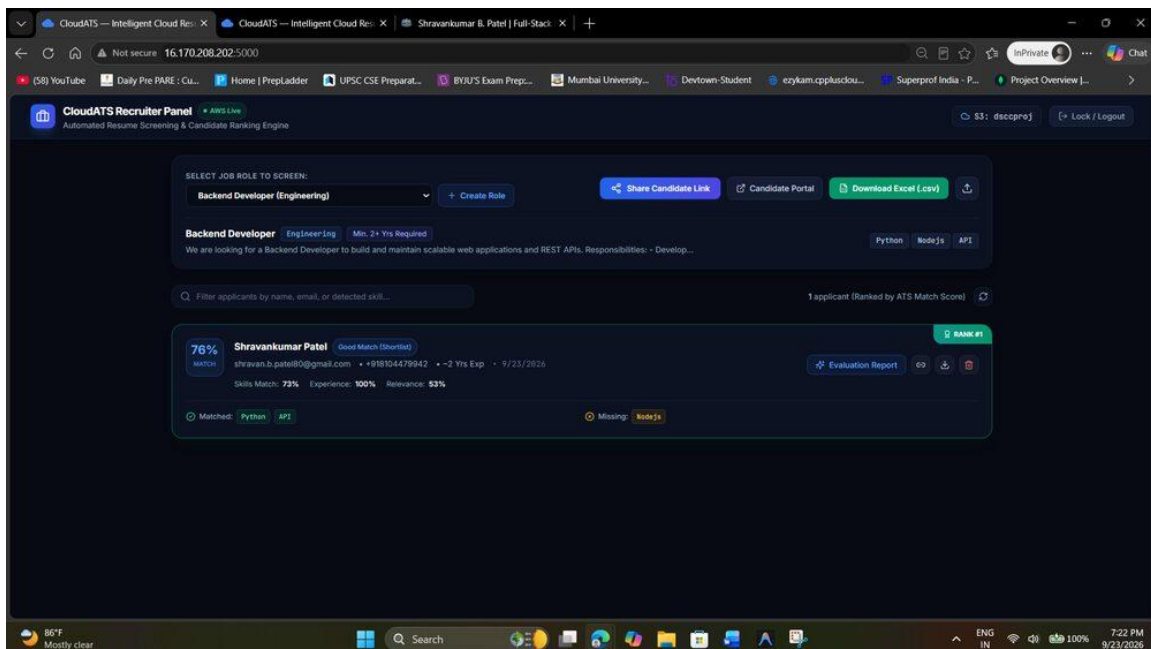


Fig 7.11a — Recruiter pipeline panel with ranked, screened candidates.

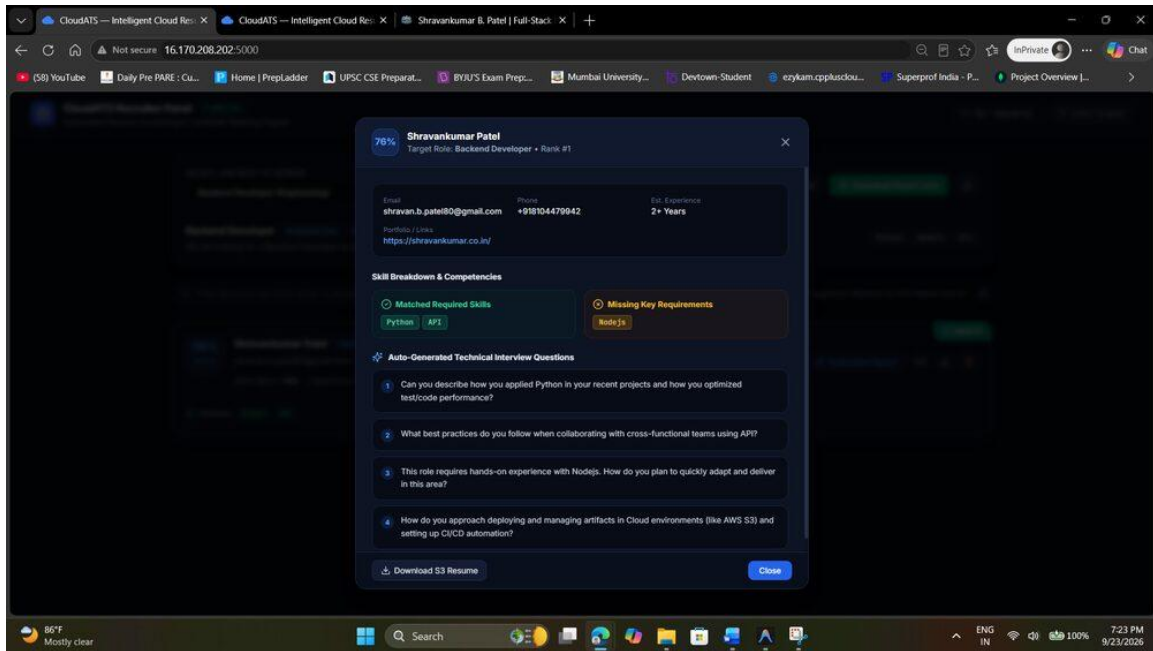


Fig 7.11b — Candidate evaluation detail panel with score breakdown.

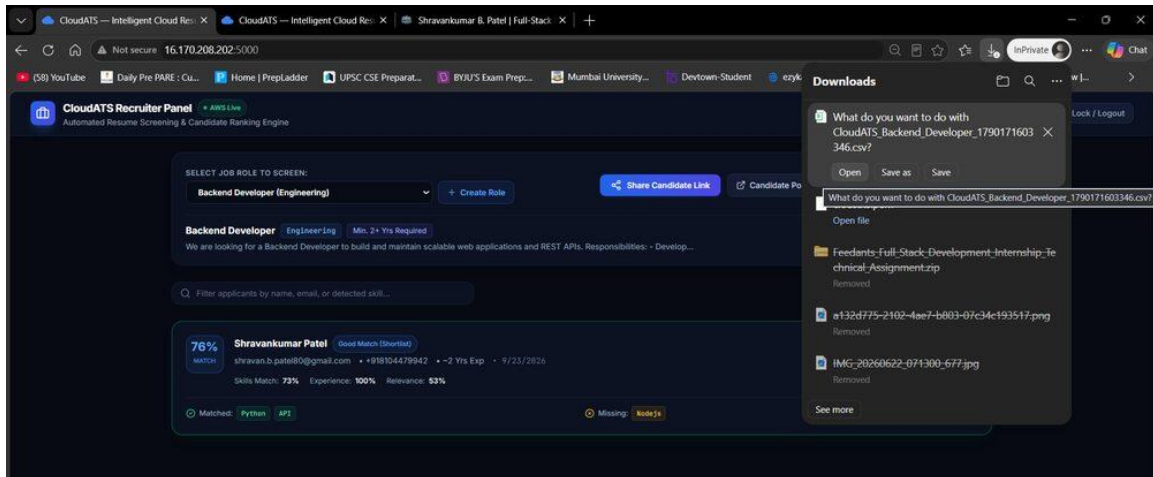


Fig 7.11c — AI-generated interview questions and candidate insights panel.

| Rank | Candidate Name     | Email                       | Phone         | Experience | Target Job        | ATS Match % | Recommendation | Matched Skills | Missing Skills | Skills Score | Experience | Relevance | Application | S3 Object Key | Notes / Portfolio           |
|------|--------------------|-----------------------------|---------------|------------|-------------------|-------------|----------------|----------------|----------------|--------------|------------|-----------|-------------|---------------|-----------------------------|
| 1    | Shrivankumar Patel | shravan.b.patel80@gmail.com | +918104479942 | 2+ Yrs     | Backend Developer | 76%         | Good Match     | Python, API    | Node.js        | 73           | 100        | 53        | 9/23/2026   | resumes/cv    | https://shrivankumar.co.in/ |

Fig 7.11d — Candidate pipeline exported to an Excel/CSV spreadsheet.

## 8. Live Deployment & Repository Links

The CloudATS platform has been fully deployed on an AWS EC2 instance and is publicly accessible. The complete source code is version-controlled and hosted on GitHub.

**Live Application:** <http://16.170.208.202:5000/>

**GitHub Repository:** <https://github.com/shravanbpatel954/CloudATS-CareerPortal>

Health Check Endpoint (server status):

```
curl http://16.170.208.202:5000/api/status
```

Expected Response:

```
{
  "status": "connected",
  "configured": true,
  "region": "eu-north-1",
  "bucket": "cloudats-resumes-bucket",
  "sesConfigured": true
}
```

## 9. Conclusion

CloudATS successfully demonstrates the design, development, and production deployment of a cloud-native Applicant Tracking System built entirely on AWS infrastructure. Through the integration of Amazon EC2, S3, Textract, Rekognition, CloudWatch, SES, STS, and IAM, the project showcases a complete real-world cloud computing pipeline — from secure resource provisioning and identity management to AI-powered document intelligence, automated notifications, and operational observability.

The project deepened the team's practical understanding of cloud architecture, AWS service integration, least-privilege security design, and full-stack deployment on a cloud virtual machine, resulting in a fully functional, live, and publicly accessible application.